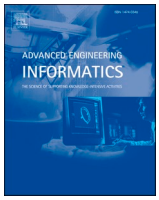
Advanced Engineering Informatics 73 (2026) 104521

Contents lists available at ScienceDirect

Advanced Engineering Informatics

journal homepage: www.elsevier.com/locate/aei



Full length article

# Automatic management of shield attitude: insights from sensor data analysis and hybrid machine learning

Hongyu Chen [a], Xingchen Zhou [b], Yong Xia [a], Yuan Cao [b], Yang Liu [c,d,*]

[a] Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Hong Kong 999077, China
[b] School of Civil and Hydraulic Engineering, Huazhong University of Science and Technology, Wuhan, Hubei 430074, China
[c] ZhongNan Hospital of Wuhan University, Wuhan University, Wuhan 430071, China
[d] Economics and Management School, Wuhan University, Wuhan 430072, China

ARTICLE INFO

*Keywords:*
Shield attitude
Attitude adaptive regulation
Sensor data
Multiobjective optimization
IBKA-NGBoost-NSGA-III
Digital twin

ABSTRACT

Shield attitude control is vital for tunnel safety, yet existing single-scene models lack generalization and real-time adaptability across diverse geologies. Herein, a digital twin (DT)-driven intelligent control framework integrating the improved black kite algorithm (IBKA), natural gradient boosting (NGBoost), and NSGA-III is proposed to achieve automated, optimal attitude management. Using a Wuhan metro underpass as a case study, the framework's performance was validated across typical geological conditions. Key results demonstrate that (1) the IBKA-NGBoost prediction model achieves an optimal accuracy-efficiency trade-off for small-sample data under two geological conditions, with R2values exceeding 0.91 for seven attitude indicators and a training time of only 29.2 s; (2) the integrated IBKA-NGBoost-NSGA-III framework enables adaptive multiscenario optimization, reducing deviations of the seven attitude parameters by an average of 69.3%; adjustment of the 12 key construction variables yielded a maximum improvement of 84.8%; and (3) embedding this algorithm into a DT system enhanced real-time regulatory capability, supporting adaptive adjustment for improved construction guidance and safety. Consequently, the proposed method enables adaptive adjustment, thereby supporting construction guidance and safety management.

## 1. Introduction

### 1.1. Research background

Metro tunnels play crucial roles in transportation and have become important infrastructure for cities [1–3]. However, during the construction process of tunnel projects, they not only face challenges from complex and variable environments but also are affected by uncertain factors such as human operations, which makes the shield position difficult to control during tunnel construction [4–7]. Furthermore, as shield tunneling technology continues to mature, the safety standards for tunnel construction have substantially increased [8,9]. Therefore, in the construction of shield tunnels, maintaining control over the position of the shield requires special attention [10]. If the shield attitude is not well controlled, a series of problems, such as rising, deviation, snake movement, displacement and cracking [11,12], which significantly affect the safety and quality of tunnel construction, can occur [13]. Moreover, several typical engineering accidents highlight the importance of controlling the attitude of the TBM [14,15]. Therefore, dynamic anticipation of the shield attitude and optimized control of the attitude deviation that is generated in the excavation route are highly important [9,16].

### 1.2. Related research

#### 1.2.1. Related research on prediction

To achieve shield tunneling attitude control, scholars have conducted extensive research. The existing methods for predicting the attitude and position of tunnel boring machines include primarily theoretical derivation, experimental methods, numerical simulation, and artificial intelligence algorithms. In terms of theoretical formulas, Liu et al. [17] integrated acceleration, momentum, and kinematic equations and azimuth equations to establish a dynamic mathematical model for TBM attitudes to achieve control of TBM trajectory tracking. Shen et al. [18] used equivalent springs to model the interplay between shield tunneling and soil, which resulted in a method for calculating the shield attitude. Huang et al. [19] introduced shield attitude into the

* Corresponding author at: ZhongNan Hospital of Wuhan University, Wuhan University, Wuhan 430071, China.
*E-mail addresses:* hongychen@polyu.edu.hk (H. Chen), zxc202110@163.com (X. Zhou), cexia@polyu.edu.hk (Y. Xia), dabailiu@whu.edu.cn (Y. Liu).

https://doi.org/10.1016/j.aei.2026.104521
Received 12 August 2025; Received in revised form 2 February 2026; Accepted 25 February 2026
Available online 14 March 2026
1474-0346/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

**Nomenclature**

| | |
|---|---|
| BKA | Black kite algorithm |
| NGBoost | Natural gradient boosting |
| NSGA-III | Nondominated sorting genetic algorithm, the third version |
| ML | Machine learning |
| BPNN | Back propagation neural network |
| RF | Random forest |
| LSTM | Long short-term memory |
| GBDT | Gradient boosting decision tree |
| SVM | Support vector machine |
| GCN | Graph convolutional networks |
| GRU | Gated recurrent unit |
| WT | Wavelet transform noise filter |
| CNN | Convolutional neural network feature extractor |
| XGBoost | EXtreme gradient boosting |
| HDSI | Horizontal deviation of the shield machine incision, mm |
| VDSI | Vertical deviation of the shield machine incision, mm |
| HDST | Horizontal deviation of the shield tail, mm |
| VDST | Vertical deviation of the shield tail, mm |
| HAA | Horizontal attitude angle, mm/m |
| PAA | Pitch attitude angle, mm/m |
| RAA | Roll attitude angle, mm/m |

horizontal force balance analysis of shield tunneling and established a theoretical model for the horizontal attitude of shield tunneling. With respect to experimental methods, Wang et al. [20] analyzed and simulated the effects of attitude and position adjustment and control through large-scale model experiments. As examples of numerical simulation methods, Shen et al. [21] studied excavation mechanics and limit ranges during the shield attitude adjustment process, and Sun et al. [22] examined how various soil parameters and burial depths affect the correction torque and shield attitude. They suggested employing attitude adjustment methods by controlling tunnel parameters. However, the factors that influence the movement behavior of tunnel boring machines are complex and varied, with coupling relationships between them. Considering real geological conditions and the complex factors that influence shields in theoretical and experimental methods is difficult. Predictions that are obtained under simplified conditions cannot provide operators with accurate guidance and are costly. Numerical simulation methods involve long modeling cycles, and the boundary conditions and parameter values are highly dependent on experience, which makes updating them in real time challenging during tunneling. Additionally, forecasting the orientation of a TBM is a real-time and dynamic highly nonlinear challenge that encompasses numerous interrelated input and output datasets. Traditional analysis methods often lack adaptability to typical geological conditions and provide insufficient real-time prediction capabilities and accuracy in cases with complex nonlinear engineering issues [15]. To overcome these limitations, data-driven approaches have emerged as a powerful alternative, shifting the focus from rigid theoretical derivations to adaptive learning from massive engineering datasets.

With advances in artificial intelligence, machine learning (ML) methods provide more innovative, scientific, and precise solutions for solving complex nonlinear engineering problems [23]. They have been extensively applied in engineering research for prediction, analysis, and optimization [24–26]. Liu et al. [27] integrated excavation efficiency and cost objectives to investigate the regulation of TBM parameters using an improved backpropagation neural network (BPNN) algorithm. Pan et al. [28] applied the support vector machine (SVM) algorithm to predict surface subsidence in conjunction with the Hongmei South Road Tunnel West Line project. SVMs are good at handling similar features, but their ability to deal with complex features and data gaps is limited. The BPNN has low efficiency and poor generalization ability. Ke Lei et al. [29] utilized the Transformer model to achieve rock classification in TBM tunnels. While deep learning architectures, particularly Transformer-based models, have gained prominence in time-series forecasting due to their self-attention mechanisms [30,31], they often require massive datasets to achieve convergence and avoid overfitting. In shield tunneling projects, high-quality labeled data for specific geological strata often exhibit “small-sample” characteristics, which limits the efficacy of data-hungry Transformer-based models. Furthermore, the high computational demand of Transformers can hinder the real-time feedback required for DTs. The natural gradient boosting (NGBoost) algorithm is a probability-based boosting algorithm that was proposed by ML researchers Tony Duan et al. in 2019 [32]. It can address challenges such as nonlinearity, data noise, and complex dependence relationships [33]. Several scholars have applied it for prediction in the engineering field [34–36]. Liu constructed a BO-NGBoost prediction model for predicting the deformation of existing tunnels [37]. Zhou proposed a BO-NGBoost prediction model that outputs corresponding safety control strategies on the basis of the safety risk level of the shield approaching the structure [38]. Cao et al. used the NGBoost algorithm to accurately predict the optimal amount of supplementary cementitious material, which served as a supportive tool for reducing the production costs and carbon emissions of segment concrete [33]. By employing the NGBoost technique, this method iteratively refines the parameter probability distributions, thereby facilitating accurate probabilistic estimation of the total thrust and torque [39]. They reported that the probabilistic prediction model based on NGBoost exhibited superior predictive performance compared with conventional deterministic ML models with the same quantity of or fewer data. In complex engineering systems and high-dimensional feature spaces, to explore patterns, a large number of data samples are usually required [40]. However, herein, the number of available samples is only approximately 1,000. The small sample size in this study is relative to tens of thousands or even hundreds of thousands of data records in some cases. As a probabilistic gradient-boosting model, NGBoost can increase robustness against both ground disturbances and equipment errors [41]. Building on previous research and analysis, in this study, the NGBoost algorithm is selected to model the intricate nonlinear relationships between the shield orientation and excavation-related parameters. However, the performance of such boosting models is highly sensitive to their configuration, necessitating a robust optimization strategy to ensure global convergence in complex search spaces.

Hyperparameters significantly influence the accuracy of NGBoost model predictions. To optimize these hyperparameters, techniques such as grid search and random search are frequently used to identify the best combination [42–44]. Many scholars have proposed the use of genetic algorithms and particle swarm optimization for hyperparameter selection. However, because both are population-based optimization techniques, they necessitate an adequate number of initial training points for effectively searching for optimal solutions, which can lead to lower training efficiency [45]. To address these constraints, IBKA is an efficient algorithm that can discover the optimal solution with the fewest objective function evaluations, thereby greatly promoting convergence speed and population diversity. It significantly outperforms all the currently popular competitive optimizers on most test functions. IBKA employs chaotic initialization and opposition-based learning to generate a highly diverse initial population that covers the solution space, thus avoiding the local-optimum traps that are caused by traditional random initialization. This strengthens the global exploration of the shield spatial–temporal search space and provides a better starting point for determining NGBoost hyperparameters such as the learning rate and

tree depth [46]. The IBKA-NGBoost model provides accurate prediction for the shield attitude and offers a basis for determining the optimal attitude.

#### 1.2.2. Related research on multiobjective optimization

Constructing the multiobjective optimization (MOO) decision model to find the optimal attitude is the next key issue. To effectively solve multidimensional objective optimization problems, Deb [47] introduced a widely distributed reference point instead of a crowding degree on the basis of the NSGA-II framework, which resulted in a new generation algorithm, NSGA-III. By preserving the key features and benefits of the NSGA-II algorithm, the NSGA-III algorithm increases population diversity and improves the ability to discover superior Pareto-optimal solutions in multiobjective scenarios, which makes it particularly adept at addressing high-dimensional multiobjective problems [48,49]. Wen Xu et al. [12] used BO-LGBM-NSGA-III to achieve MOO for surface subsidence, cutterhead wear, and shield operation energy consumption. Chen Hongyu et al. [3] used BO-CatBoost-NSGA-III to achieve dynamic MOO for tunneling energy, speed and tool wear. NSGA-III predefines a set of uniformly distributed reference points to steer the search toward diverse trade-offs on the Pareto front and keep the algorithm from becoming trapped in local optima [50]. However, at present, research on the optimization of shield parameters and attitude is mainly focused on the algorithms used, with limited attention given to the integration of optimization models and physical entities [51].Therefore, it is necessary to establish a framework that can not only accurately predict attitude but also convert modeling insights into reliable automated control strategies.

The emergence of digital twin (DT) technology is regarded as a promising approach to bridging the aforementioned issues. DT technology enables interactions among simulation, optimization, and feedback, and this capability creates new opportunities for intelligent control in tunnel construction [52]. Herein, a new method combining the IBKA-NGBoost-NSGA-III intelligent hybrid algorithm with the DT platform is proposed to promote intelligent adaptive attitude adjustment. Currently, research is mainly focused on the application of artificial intelligence methods in specific geological settings [7], and emphasis is placed on the application of construction parameters based on digital twins [40]; however, research on the application of DTs for attitude control and evaluation is relatively scarce. This study combines MOO and a DT to explore the optimization and control of shield machine attitude under different geological conditions, and the implementation mechanism of the DT is systematically evaluated to promote DT deployment in the shield construction safety field.

### 1.3. Gaps and contributions of this study

#### 1.3.1. Research gaps

Previous traditional methods and data-driven research have improved the reliability and efficiency of tunnel excavation. This is substantive progress in this field, but specific constraints have also been identified:

(1) Emphasis on individual geological conditions: Prior research has often focused on specific types of geological conditions and has not succeeded in developing a universal model that is applicable to multiple geological scenarios. A comprehensive strategy is essential for addressing the complex geological challenges in tunnel engineering.
(2) Methodological variation: There is still no consensus regarding the adoption of deep learning methods that are related to shield spatial–temporal analysis or the application of ensemble algorithms, which depend on the sample size and research objectives. The differences among these methods may affect the consistency and validity of the research results.
(3) Lack of adaptive optimization regulation and evaluation mechanisms: The current data-driven methods use existing data for prediction, which results in a lag effect. The objective of the engineering problem of determining the shield attitude should focus on optimization rather than prediction. Prediction is merely the logic for achieving the objective of control. Therefore, adaptive decision regulation for MOO and a DT evaluation mechanism are needed to effectively develop real-time intelligent DTs.
(4) Underused technological progress: In previous research, sensor data, ML, the Internet of Things, and other emerging technologies have rarely been used to develop an intelligent decision-making framework for the adaptive attitudes of shield tunnels using DTs. Integrating these technologies can increase the efficiency, precision and initiative of decision-making. This study focuses primarily on investigating the predictive regulation of a single geological condition rather than adaptive optimization decision-making [14].

#### 1.3.2. Motivation and contributions

During shield tunneling, the attitude of the machine easily becomes unstable and deviates under dynamically changing, complex geological conditions; thus, rapid response and timely control adjustments are essential. Moreover, as many as seven objectives must be simultaneously considered for attitude regulation. Therefore, determining which method to adopt and how to bridge the resulting research gaps are key issues that must be addressed. On the basis of the research status and problem-solving methods analyzed, the fundamental research inquiries of this study focus on the following:

(1) How can an intelligent prediction framework be constructed without a large amount of sample data to accurately describe the complex nonlinear relationship between the shield attitude and influencing factors?
(2) How can the significance of parameters be determined and assessed, how can the key control parameters that influence the shield orientation during construction be identified, and how can effective control of the shield attitude be achieved?
(3) How can MOO be conducted on the basis of key parameters to achieve real-time dynamic adaptive control of the shield attitude under varying geological conditions?

Given the above problem analysis, herein, a DT platform with an intelligent IBKA-NGBoost-NSGA-III optimization control framework for the tunneling attitude is presented. The aim of this study is to provide a feasible method for adaptive real-time intelligent dynamic control of the shield attitude. To address the aforementioned gaps, the following contributions are made:

- A novel DT-driven framework for adaptive shield attitude control: A closed-loop intelligent framework is proposed by integrating IBKA-NGBoost and NSGA-III, enabling real-time dynamic interaction between physical tunneling entities and virtual models to achieve automated attitude management across varying geological conditions.
- High-efficiency predictive modeling via IBKA-NGBoost under small-sample constraints: By optimizing the NGBoost algorithm with the IBKA, the model achieves excellent uncertainty quantification and generalization. It yields high precision ($R^2 > 0.91$) with minimal training time (29.2 s), specifically addressing data scarcity issues for complex strata.
- Multiobjective collaborative optimization mechanism for complex tunneling parameters: The coupling of the intelligent “decision engine” (NSGA-III) with the prediction model allows for the simultaneous optimization of 12 critical construction variables. This mechanism achieves a maximum improvement of 84.8% in attitude

accuracy compared to that for baseline data, significantly reducing reliance on operator experience.

- Paradigm shift from passive "AI-prediction" to proactive "AI-decision making": The study establishes a collaborative decision-making protocol between human operators and intelligent models. By embedding the TOPSIS evaluation method into the DT platform, we transform a traditional AI method from a simple predictive tool into a robust autonomous control system for TBM operations.

### *1.4. Organization structure*

The remainder of this paper is organized as follows: Section 2 outlines the methodologies that are employed in this study. Section 3 introduces the analysis and decision-making methods and reports the use of a metro underpass project in Wuhan as a case study to perform multiobjective excavation parameter prediction and optimization control. Section 4 discusses the advantages of the IBKA-NGBoost model over other models, its DT framework and future work. Finally, Section 5 offers a comprehensive overview of the study's findings.

## 2. Methodology

To address the high-dimensional MOO problem related to the attitude of the TBM and achieve intelligent decision-making in the tunneling system, in this study, a high-dimensional MOO method for the shield attitude that is based on the DT platform integrated with the IBKA-NGBoost-NSGA-III algorithm is developed. The primary process is outlined as follows: First, a framework of the shield attitude DT system is established, and the physical entity is mapped to a virtual model. On this basis, data are obtained through the DT platform to offer numerical sources for intelligent decision-making. Second, intelligent prediction is achieved through the integration of intelligent modules. Finally, control from the virtual entity to the entity is achieved, and the shield attitude is controlled. Moreover, the performance of the TBM is improved. The intelligent online real-time adaptive optimization and decision-making process is displayed in Fig. 1.

### *2.1. Establishment of the DT model for shield attitude control*

#### *2.1.1. DT framework*

Attitude control in shield tunneling using DTs is a stage in which shield tunneling is simulated in a virtual construction environment. Herein, DTs are introduced into shield tunneling attitude analysis, and an intelligent shield tunneling attitude analysis framework that is via DTs is proposed, which provides a visual decision-making environment for controlling the orientation of a TBM. The framework of DTs, which includes the following main parts, is shown in Fig. 2.

1. System Architecture and Components

(1) Physical layer

The physical layer consists of the physical entities, surrounding construction facilities, data collection sensors, and activities of construction personnel. Its main functions include completing the basic construction tasks of the shield machine, providing parameters for twin body modeling, and collecting and transmitting data. To obtain various parameters, the shield machine is equipped with different types of sensors. For instance, the soil chamber pressure is monitored by pressure sensors installed at different positions (upper part, lower part, left side, and right side) of the soil chamber. The corresponding variables are: X1 (upper part), X2 (lower part), X3 (left side), and X4 (right side). The comprehensive automated monitoring system can monitor and record the various attitude parameters of the shield machine in real time, including all the attitude parameters in this study. Additionally, the data acquisition frequency is set to 30 s per collection sequence and the Industrial Ethernet data communication protocol is used to ensure the reliable acquisition of real-time data.

(2) Digital layer

This layer includes virtual models of all the physical layer components and maps the behavior of the physical layer. This involves establishing models for the appearance and rules of physical systems, including prediction and optimization algorithms, along with controlling physical shield tunneling construction and managing project engineering-related information. The construction process of the virtual model includes 3D geometric modeling of the shield machine based on Unity 3D, as well as modeling of behavior rules. The IBKA-NGBoost-NSGA-III algorithm is encapsulated as a callable microservice and embedded into the digital thread.

(3) User layer

The user domain of DTs includes various participants who can make

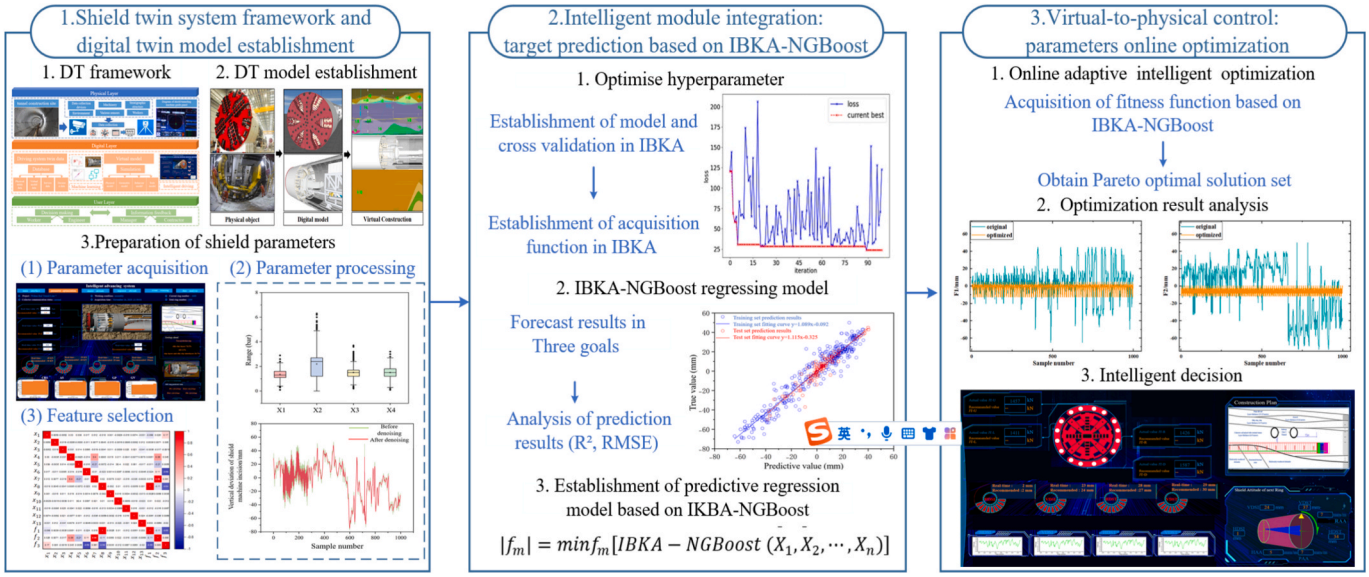



**Fig. 1.** Flowchart of the optimization of shield attitude on the basis of IBKA-NGBoost-NSGA-III and DTs.

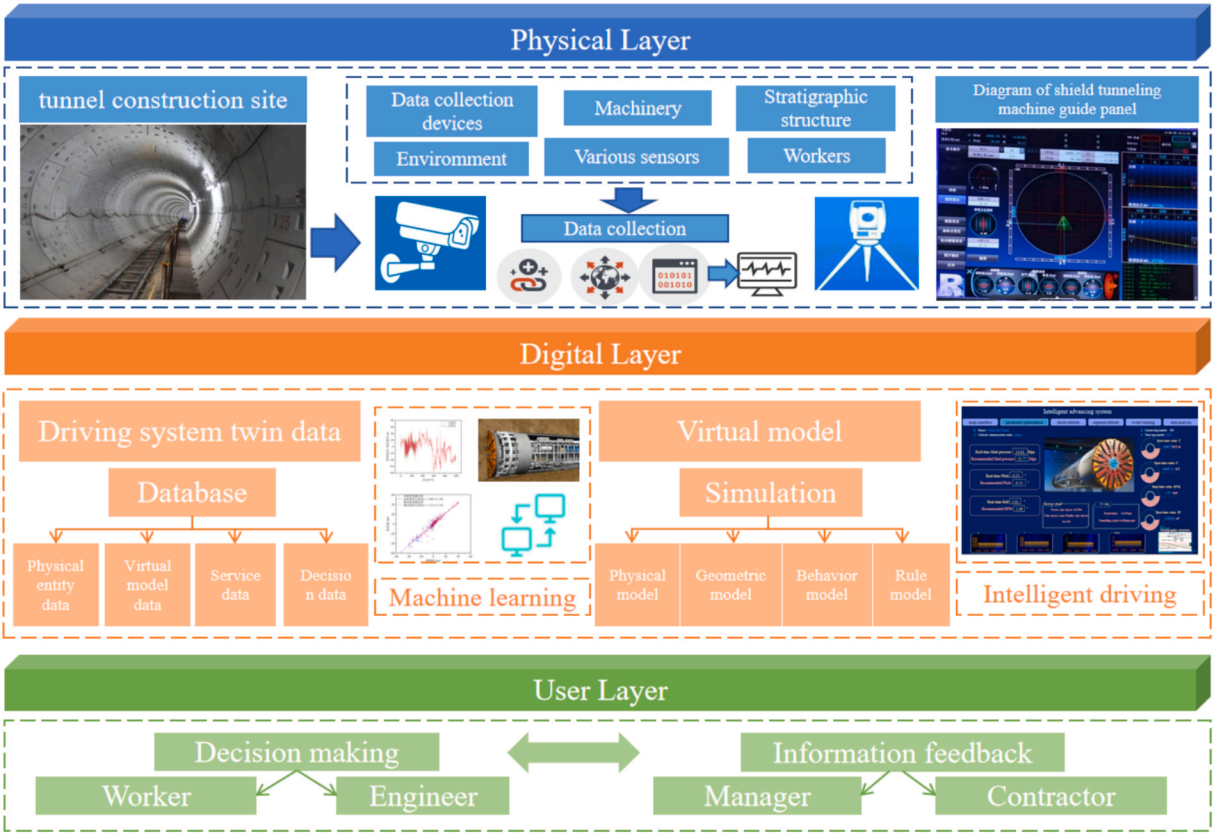



**Fig. 2.** Shield tunneling DT system framework.

information-based decisions and provide feedback. A web-based visual dashboard enables engineers to monitor parameters in real time and receive optimization suggestions. The system presents the optimized results calculated by the digital layer (i.e., the recommended construction parameters and attitude parameters) to the user clearly, as shown in Fig. 15. Simultaneously, the "actual values" and "recommended values" of the parameters are displayed in the interface, providing a basis for the user's real-time decision-making.

2. Data flow and integration mechanism

The physical and virtual entities are bidirectionally mapped and corresponding data are synchronized through the Internet of Things. The collected data are provided externally through OPCUA as standardized data services. This specifically includes the data preprocessing process, the selection of real-time databases, and a closed-loop path for optimizing instructions to be issued from the virtual space to the physical shield machine control system.

*2.1.2. Physical-to-virtual reflection: Establishment of a DT model*

The sensors in the TBM persistently track and record the operating status throughout the excavation process, and the data are processed by the acquisition system. These sensors capture important information such as the shield excavation time, construction progress, construction parameters, and shield attitude. At present, the attitude and construction parameters are displayed on the screen in the control room of the TBM and can be remotely viewed for monitoring. However, the lack of intuitive real-time dynamic visualization functions hinders real-time analysis of the machine's operating status and safety, which results in an inability to fully understand the substantial volume of data generated in the TBM excavation process. To address this challenge, a DT platform was created as an effective means for reflecting physical entities. The mapping between the shield physical object and the virtual representation is shown in Fig. 3.

*2.1.3. Operational mechanism of the DT system*

Based on the aforementioned DT platform, real-time intelligent control of the shield attitude is carried out. The operational mechanism of the DT system is as follows.

Step 1: Real-time Sensing. Multisource operational and attitude data are synchronously acquired in the physical layer using TBM-embedded sensors, including earth pressure cells, thrust jack transducers, and the guidance system. Following preliminary noise reduction via the local PLC, data packets are uploaded to the DT database at 1-second intervals (1 Hz) through an industrial Ethernet backbone, establishing a robust high-fidelity nexus between the physical and digital domains.

Step 2: Intelligent Prediction. Serving as the "intelligent brain", the digital layer first performs preprocessing on the incoming streaming data. The system then invokes the pretrained IBKA-NGBoost algorithm module, which not only predicts the current attitude deviation based on historical tunneling data but also provides an uncertainty estimation of the prediction through its probabilistic output characteristics. Compared to traditional point-estimate prediction methods, this probabilistic paradigm aids in identifying latent risks and establishing a more scientific basis for threshold judgment. An early warning is triggered when the predicted deviation probability exceeds the 80% confidence limit.

Step 3: Intelligent Optimization. Upon completion of the prediction, the system implements real-time MOO using the IBKA-NGBoost-NSGA-III framework. The optimization primarily aims to minimize attitude deviations while simultaneously enhancing the tunneling efficiency of the TBM process. By exploring the predefined search space of operational parameters, the algorithm identifies a set of Pareto-optimal solutions. The digital layer then filters the most compatible recommended parameters from the Pareto set, tailored to the current geological conditions, to serve as the basis for correction.

Step 4: Decision Feedback. The final optimization recommendations

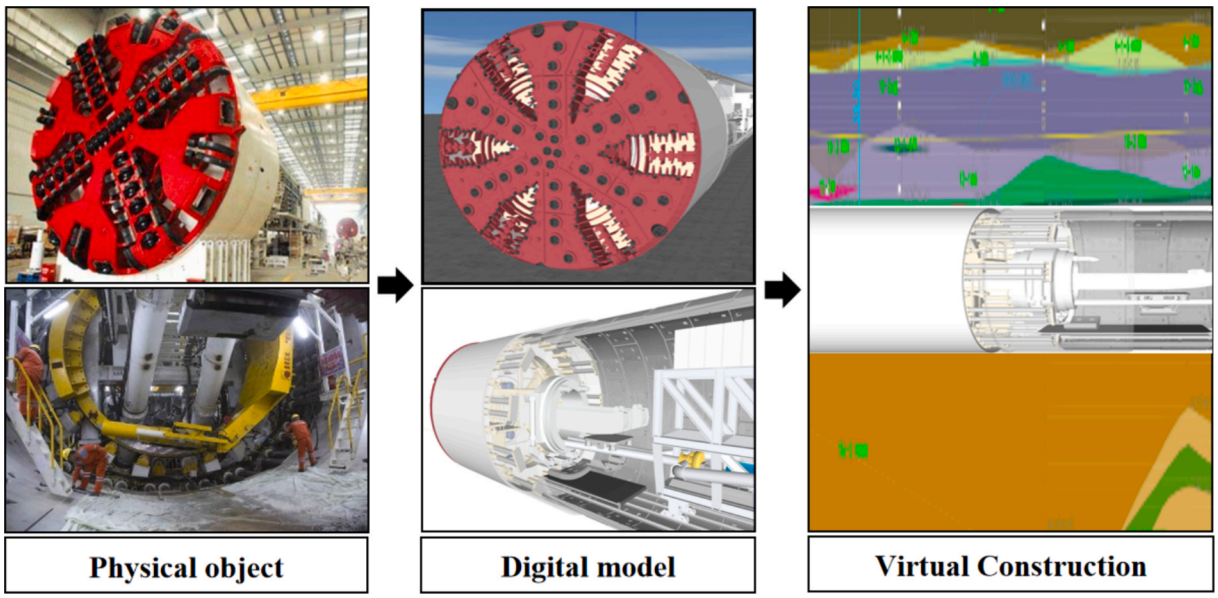



**Fig. 3.** Mapping from physical entities to virtual entities.

are transmitted to the TBM operator via the web-based User Layer. Through 3D visualization technology, the interface dynamically displays the TBM's real-time position and predicted trajectory within the tunnel, presenting a comparative visualization of "current parameters" versus "system-recommended parameters". Operators can adopt the suggestions with a single click or make fine-tuned adjustments based on field experience. Once the instructions are executed by the physical hardware, the resulting feedback data are looped back to Step 1, forming a robust "perception-analysis-optimization-feedback" closed loop for real-time dynamic supervision of shield attitude.

#### *2.1.4. Feature selection and sample preprocessing*

1. Parameter variable selection

Herein, four shield machine position indicators are selected as the output variables of the model: the horizontal deviation of the shield machine incision (HDSI), the vertical deviation of the shield machine incision (VDSI), the horizontal deviation of the shield tail (HDST), and the vertical deviation of the shield tail (VDST). Additionally, three angle indicators are chosen: the pitch attitude angle (PAA), the rolling attitude angle (RAA), and the horizontal attitude angle (HAA). The HDSI represents the horizontal distance between the center of the TBM cutterhead and the design tunnel axis (DTA) at a given moment. A positive deviation is recorded when the cutterhead center deviates to the right of the DTA. Similarly, the VDSI indicates the vertical distance between the nucleus of the cutterhead and the design tunnel axis (DTA), where a positive deviation signifies an upward shift toward the DTA. The same logic applies to the HDST and VDST. The HAA, which is also known as the yaw angle, is the deflection angle of the TBM in the horizontal direction and describes the variation in the tunneling route in the horizontal direction. This angle requires frequent correction control. The PAA represents the vertical attitude of the shield machine, specifically its deflection angle while tunneling vertically. Owing to its own gravity, the "top-heavy and bottom-light" phenomenon of the shield machine bowing downward can occur, which necessitates precise control of the PAA variation. The RAA is the rotation angle of the TBM around its axis because of the force from the cutterhead's rotation during propulsion. The RAA is defined as positive when the TBM rotates clockwise around its axis. To ensure normal forward tunneling of the shield machine, the RAA must be kept within a small range. As tunneling progresses, the attitude and position deviate from the DTA, and the seven considered parameters also change continuously. To accurately predict and control the shield position and attitude, precise measurements of these attitude indicators are needed [15,53]. Geological parameters and shield construction factors are the key factors that influence the attitude and positioning of the shield machine. Owing to the short excavation section under the tunnel, the burial depth and geometric shape of the tunnel remain almost unchanged, and their effects can be ignored. On the basis of expert experience and the literature [9,14,15], in this study, the soil warehouse pressure—upper, soil warehouse pressure—lower, soil warehouse pressure—left, soil warehouse pressure—right, threshold difference between upper and lower jacks, left and right jack thrust difference, travel difference, articulation cylinder stroke difference (horizontal), articulation cylinder stroke difference (vertical), articulation oil pressure, cutterhead rotation speed, cutterhead torque, advancing speed, grouting pressure, grouting volume, spiral rotation speed, screw torque, internal friction angle, soil cohesion, compression modulus, and other input parameters are selected.

2. Data preprocessing

(1) Data Cleaning

During the TBM shutdown process, a substantial volume of blank data with parameter values of 0 are generated. To create a dataset, eliminating any blank data to increase the data validity is essential. The data cleaning process is implemented based on Equations (1) and (2) [54,55]:

$$f = f(v)\cdot f(n)\cdot f(T)\cdot f(n_1)\cdot f(T_1) \tag{1}$$

$$f = \begin{cases} 1, & Operational\,state \\ 0, & Non-operational\,state \end{cases} \tag{2}$$

where $v, n, T, n_1,$ and $T_1$ represent the advancing speed, cutterhead rotation speed, cutterhead torque, screw conveyor rotation speed and screw torque, respectively. If $f = 0$, the entry is deleted.

(2) Box plot method

By displaying the distribution of data, a box plot effectively highlights outliers within the dataset, with values that fall outside the designated upper and lower boundaries classified as outliers. The box

plot method is used to screen outlier points in the sample data to prepare for the subsequent processing and correction steps.

(3) Data normalization

Because there are significant differences in the ranges and magnitudes of the input variables, as well as positive and negative differences, directly using them as training samples increases the error of the prediction model. Therefore, normalizing the original data is necessary. Standardizing the data to the [-1, 1] interval is a more effective approach for preventing the suppression of input vector features than standardizing to the [1] interval is. The formula for normalization is as follows [50,56]:

$$x_{nor} = (\overline{x}_{max} - \overline{x}_{min}) \cdot \frac{x - x_{min}}{x_{max} - x_{min}} + \overline{x}_{min} \tag{3}$$

where $x_{nor}$ represents the normalized standard value; $\overline{x}_{max}$ and $\overline{x}_{min}$ default to 1 and −1, respectively; $x$ represents the sample value; and $x_{max}$ and $x_{min}$ represent the maximum and minimum sample values, respectively.

### 2.2. Shield tunnel attitude prediction using the IBKA-NGBoost model and DTs

Through the use of real-time monitoring data from the TBM, DTs can be combined with an intelligent prediction model to accurately estimate the attitude index. This includes building and training the IBKA-NGBoost learning model. However, selecting an appropriate model architecture is critical given the complexity of the tunneling environment and the specific characteristics of the sensor data.

#### 2.2.1. Improved black kite algorithm (IBKA)

The BKA addresses optimization issues by mimicking how black kites hunt and capture prey and uses both global and local search methods to identify the best solution [57]. The IBKA maintains the strengths of the original BKA and introduces a population strategy that is grounded in chaotic opposition. This helps improve the quality of the population and its distribution probability at the beginning. Moreover, IBKA employs the Levy flight approach and a nonlinear decreasing inertia weight approach to strike a balance between the exploration and exploitation abilities of the whole population, all while remaining in line with the natural movement patterns of black kites [58]. The formulas for calculating the predatory stage of the BKA are presented below:

$$X_i = BK_{lb} + rand(BK_{ub} - BK_{lb}) \tag{4}$$

$$x_{t+1}^{i,j} = \begin{cases} x_t^{i,j} + n(1 + sin(r)) \times x_t^{i,j}, 0.9 < r \\ x_t^{i,j} + n \times (2r - 1) \times x_t^{i,j}, else \end{cases} \tag{5}$$

where $BK_{lb}$ and $BK_{ub}$ represent the upper and lower limits, respectively, of the threshold; rand represents a random value in the range of [1]; $x_t^{i,j}$ and $x_{t+1}^{i,j}$ represent the positions of the $i$-th black kite in the $j$-th dimension at iterations t and t + 1, respectively; r represents a random value between 0 and 1; T denotes the total number of iterations; and $n = 0.05 \times exp\left(-2 \times (t/T)^2\right)$.

Following the incorporation of the Levy flight escape strategy and the nonlinear decreasing inertial weight strategy for enhancement, the formulas for updating positions in the predatory and migratory stages of IBKA can be expressed as follows:

$$X_{i,j}(t+1) = \begin{cases} Lv \times X_{i,j}(t) + w(t) \times n(1 + sin(r)) \times X_{i,j}(t) & p < r \\ Lv \times X_{i,j}(t) + w(t) \times n(2r - 1) \times X_{i,j}(t) & else \end{cases} \tag{6}$$

$$X_{i,j}(t+1) = \begin{cases} Lv \times X_{i,j}(t) + C(0,1) \times \left(Lv \times X_{i,j}(t) - w(t) \times BW_L(t)\right) \\ f\left(X_{i,j}(t)\right) < rf\left(X_j(t)\right) \\ Lv \times X_{i,j}(t) + C(0,1) \times \left(w(t) \times BW_L(t) - Lv^* m \times X_{i,j}(t)\right), else \end{cases} \tag{7}$$

To increase the accuracy of the globally optimal value that is obtained by integrating the Levy flight strategy with the nonlinear decreasing inertia weight strategy, a greedy strategy is employed for position updating in this study. This strategy only performs replacement when the fitness of the new position is superior to that of the original position, which ensures that a better individual is selected in each update. Otherwise, the position of the solution remains unchanged.

#### 2.2.2. NGBoost

NGBoost is a boosting algorithm that focuses on probabilistic predictions and enables direct estimation of the probability distribution for the target variable. This method quantifies the uncertainty in the prediction results and provides uncertainty assessments for point predictions. Additionally, NGBoost employs the natural gradient descent method for model training, which can converge to the globally optimal solution faster [41,59].

NGBoost comprises three primary modules: (1) base learners ($f^n$), (2) a parameter probability distribution ($P_\theta(y|x)$), and (3) reasonable scoring rules ($\wp(P_\theta, y)$). For a new sample input $x$, the prediction $y|x$ is obtained in the form of $P_\theta$. $\theta$ is a combination of $m$ base learner inputs and an initial distribution $\theta^{(0)}$, and the above parameters determine the probability prediction $y|x$. To obtain the prediction result parameter $\theta$ for a certain $x$, each base learner $f^n$ takes $x$ as input and, on the basis of a scaling factor $\rho^{(m)}$ and learning rate $\eta$, scales the predicted output. Here, $f^n$ is collectively referred to as the set of base learners for stage $n$. The updated equation is as follows [60]:

$$y|x\ P_\theta(x), \theta = \theta^{(0)} - \eta \sum_{m=1}^{M} \rho^{(m)} \cdot f^{(m)}(x) \tag{8}$$

The prediction parameters for each sample $i$ are calculated for its current position in each iteration $\theta_i^{m-1}$. The natural gradient $g_i$ of the evaluation rule $S$ is used to fit a set of base learners $f$ to predict the corresponding components of the natural gradient for each sample $x_i$ After the scaling factor $\rho^{(m)}$ is determined, each predicted sample parameter has a proportional gradient that is based on a smaller learning rate $\rho^{(m)} \cdot f^{(m)}(x)$; after scaling is performed, it is added to each $\theta_i^{(m-1)}$ and updated to $\theta_i^m$.

NGBoost uses decision trees as base learners and updates the prediction model parameters through iterative computation of the natural gradient. This model undergoes an iterative training loop, which includes natural gradient computation for optimization, fitting of base learners, and adaptive model updates. This enables the model to output the full probability distribution of the target variable. Upon comparison, the normal distribution shows optimal performance in terms of the CRPS and PIT calibrations and is thus selected as the target distribution.

The hyperparameters of the NGBoost model are optimized using IBKA, with a focus on four key parameters: learning_rate, minibatch_frac, col_sample, and n_estimators. In gradient boosting trees, the learning_rate dictates the magnitude of weight adjustments in each iteration of the model's training process. An excessively high learning rate may lead to unstable training, whereas an overly low learning rate may result in a sluggish training process. Minibatch_frac typically refers to the proportion of the data subset that is used to train the model in each iteration. During the tree construction process, col_sample randomly selects a certain proportion of features for splitting, which helps prevent overfitting and improves model diversity. N_estimators refers to the number of trees. Increasing the number of trees can improve model performance but also increases computational costs and the risk of overfitting. The IBKA algorithm is configured for 100 iterations with a population size of 5. The dataset is randomly divided into two parts:

80% for training and 20% for testing. Additionally, a fivefold cross-validation method is employed to mitigate the effects of random variance and prevent overfitting.

#### 2.2.3. Accuracy testing of the prediction models

To achieve a more precise quantitative assessment of the effectiveness of IBKA-NGBoost [61], the root mean square error (RMSE) and the mean absolute error (MAE) are used to assess the extent of inconsistency between the predicted and actual values [62]. $R^2$ is used to assess the degree of correspondence between the predicted and actual values [63]. The calculation method is shown in Eqs. (9) to (11) [64].

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2} \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \tag{10}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\widehat{y}_i - y_i| \tag{11}$$

### 2.3. Intelligent regulation from virtual to real: Online parameter optimization

#### 2.3.1. NSGA-III

The NSGA-III algorithm increases the search rate for high-quality solutions, maintains population diversity, and helps the algorithm effectively explore Pareto frontiers in multidimensional optimization. Research indicates that the NSGA-III algorithm excels at solving problems with three or more objectives [50,65]. It uses reference points to divide the population's solutions and calculates the crowding distance on the basis of these reference points. By adjusting the number and positions of the reference points, the distribution of solutions can be adjusted to better cover the front edge [66].

Each reference point that is constructed by NSGA-III is on an $(M-1)$-dimensional hyperplane, where $M$ represents the dimension of the objective space, i.e., the number of optimization objectives. If we divide each target into $H$ parts, then the number of reference points is [49]:

$$\text{ref Count} = C_{M+H-1}^{H} \tag{12}$$

Next, NSGA-III normalizes the objective functions. First, the minimum values of each objective function are combined and set as ideal points [67]:

$$\overline{z} = \left(Z_1^{min}, Z_2^{min}, \cdots, Z_M^{min}\right) \tag{13}$$

In the formula, $Z_1^{\min}, Z_2^{min}, \cdots, Z_M^{min}$ represent the minimum values of the optimization functions. The values of the distinct population within the target dimension are then converted as follows:

$$f_i'(X) = f_i(X) - Z_i^{min} \tag{14}$$

The extremum points of each dimension are acquired through the ASF function operation [68]:

$$ASF(X, W) = \max_{i=1} \frac{f_i'(X)}{w_i} \tag{15}$$

The lowest value of the ASF for each dimension is defined as the extremum point, a set of $M$-dimensional extremum points is obtained, and this set is used to construct a hyperplane with $(M-1)$ dimensions compared to each coordinate axis. The intercept between the hyperplane and each coordinate axis is denoted as, and the objective function can be normalized by the following equation [68]:

$$f_i^n(X) = \frac{f_i'(X)}{a_i - Z_i^{min}} = \frac{f_i(X) - Z_i^{min}}{a_i - Z_i^{min}} \tag{16}$$

The next step is to create a "mapping relationship" among the individuals in the population and the reference points. The origin of the hyperplane is connected with various reference points, rays are constructed, and individuals with the smallest distances are grouped together as related reference points. The number of individuals that are linked to each reference point is determined, and $K$ individuals are selected to enter the final frontier population on the basis of this number.

#### 2.3.2. Determining the objective function and establishing variable constraints

The relational function is expressed as follows [69]:

$$|f_m| = minf_m[IBKA - NGBoost(X_1, X_2, \cdots, X_n)] \tag{17}$$

where $|f_m|$ represents the absolute value of the output function for each optimization objective and where $X_1, X_2, \cdots, X_n$ represent the input variables.

The relevant constraint range, which is expressed as follows, is provided on the basis of engineering specifications and project experience [69]:

$$v_{bl} \le x_i \le v_{ul} \tag{18}$$

where $x_i$ represents the measured value of decision variable $X_i$ and where $v_{ul}$ and $v_{bl}$ indicate the upper and lower bounds, respectively, of the input variable $X_i$.

#### 2.3.3. TOPSIS approach for deriving the optimal solution

The Pareto-optimal solutions that are generated by NSGA-III form a collection that effectively addresses MOO requirements. However, practical tunneling operations necessitate a single decision-making strategy to guide excavation activities. When the final optimal solution is selected from the Pareto front, various multicriteria decision-making methods, such as those that are based on fuzzy logic or utility theory, are available. However, fuzzy logic requires the membership functions to be manually defined, which introduces strong subjectivity, and utility theory requires the decision-makers' utility curve to be obtained in advance, which makes real-time construction on site difficult. In this study, the TOPSIS method is adopted for the following reasons: First, the Pareto solution set that is generated in this research consists of precise numerical objectives; thus, no fuzzification is needed. Second, the "closeness to the ideal solution" concept of TOPSIS is intuitive and can provide a clear ranking of compromise alternatives. Third, TOPSIS has been widely applied to shield tunneling construction problems, and the use of TOPSIS to screen the optimal solution can offer valuable insights for similar projects [70,71]. Therefore, TOPSIS is considered the most suitable approach for addressing the decision-making problem in this study. This method evaluates each candidate by measuring its distances to both the positive (ideal) and negative (nonideal) reference points. This dual-proximity assessment enables systematic ranking of trade-off solutions under complex geological and engineering constraints. The scoring equation for each alternative is presented as Equation (19) [72]:

$$S_i = \frac{f_i^-}{f_i^- + f_i^+} \tag{19}$$

where $S_i$ denotes the fraction that is associated with the $i$-th Pareto solution, $f_i^-$ indicates the distance between the $i$-th Pareto solution and the negative ideal solution, and $f_i^+$ represents the distance between the $i$ th Pareto solution and the positive ideal solution. The calculation formulas for $f_i^-$ and $f_i^+$ are presented in Equations (20) and (21), respectively:

$$f_i^- = \sqrt{\sum_{j=1}^{m}\left(x_{ij} - z_j^-\right)^2} \tag{20}$$

$$f_i^+ = \sqrt{\sum_{j=1}^{m}\left(x_{ij} - z_j^+\right)^2} \tag{21}$$

where $m$ represents the overall count of the optimization objectives, $x_{ij}$ represents the $i$-th Pareto solution under the $j$-th objective, and $z_j^+$ and $z_j^-$ denote the highest and lowest values under the $j$-th objective, respectively. The corresponding calculation formulas are displayed in Equations (24) and (25):

$$z_j^+ = max\{x_{1j}, x_{2j}, \cdots, x_{ij}\} \tag{22}$$

$$z_j^- = min\{x_{1j}, x_{2j}, \cdots, x_{ij}\} \tag{23}$$

To showcase the degree of improvement that is attained by each optimization approach, the percentage of improvement for each Pareto solution is computed for individual objectives by Equation (24), and the total improvement that is offered by each scheme is assessed by Equation (25):

$$Imp_{ij} = \left|\frac{Org_{j,ave} - Opt_{ij}}{Org_{j,ave}}\right| \tag{24}$$

$$Impoverall_i = \sqrt{\sum_{j=1}^{m}\left(Imp_{ij}\right)^2} \tag{25}$$

where $Imp_{ij}$ represents the improvement that is attained for the $j$-th objective of the $i$-th Pareto solution, $Org_{j,ave}$ denotes the initial average of the $j$-th objective, $Opt_{ij}$ signifies the optimization value of the $j$-th objective of the $i$-th Pareto solution, and $Impoverall_i$ represents the cumulative improvement that is provided by the $i$-th Pareto solution.

#### 2.3.4. Online optimization implementation steps

Herein, the IBKA-NGBoost-NSGA-III algorithm is used to develop an optimized solution for the multiobjective challenge of determining shield tunneling parameters. The optimization process begins with a randomly initialized population, which moves through procedures of selection, recombination, and mutation to create successive generations of viable solutions. In the process of seeking high-quality solutions, iterative evolutionary strategies such as nondominated sorting, the reference point method, and associated solutions are employed to continuously generate better populations and update the space of feasible solution sets. Pseudocode that outlines these steps is presented in Fig. 4.

## 3. Case study

### 3.1. Background

In this study, intelligent control and decision-making for shield attitude in a specific section of the Wuhan Metro Tunnel in China are used as a case study. This study focuses on the underpass construction between Julong Avenue Station and Tangyunhai Station of the Wuhan Metro Line 7 project, which intersects with Line 2. At the underpass location, the existing pipeline segment features an inner diameter of 6 m and is buried at a depth of 8.5–9.92 m and situated in cohesive soil that is interspersed with crushed stones. In contrast, the pipe segment for Line 7 features an outer diameter of 6.2 m and is buried at a depth of 18.62 m and positioned within a layer of cohesive soil that is mixed with crushed stones and moderately weathered dolomite. Notably, the separation between the underpass section and the existing line is merely 2.7 m.

The construction of this tunnel project has high requirements for the attitude and position of the TBM, and new intelligent technologies for predicting the attitude of the TBM are necessary. The shield machine attitude parameters and geological environment around the tunnel are shown in Fig. 5.

### 3.2. Data acquisition and analysis

#### 3.2.1. Data acquisition

The geological data are sourced from geological exploration reports. The shield data are monitored in real time through the DT platform, and the data between rings number 1455 and 1465 of the underpass section are collected. The data collection interval is 30 s. Ultimately, 1048 datasets are selected as sample data to construct a research model. Each dataset includes 18 parameters and 3 geological parameters obtained by the real-time measurement system of the TBM. The methods that are

**Algorithm: NSGA-III optimization base on the prediction of IBKA-NGBoost**

**Input:** X: Operational parameters, Model; Meta-model for objective 0, N: Number of population

**Output:** Optimized parameters $x$; Optimized objectives $f_i$

**Construct:** Meta-model $f_i(x)$= *IBKA-NGBoost* algorithm

**Set:** the parameter of NSGA-III

**DO**

Define optimization problem: $min.f(x)=\{f_1(x), f_2(x), f_3(x) \ldots\}$

Generate initial population $P^0 = \{p_1^0, p_2^0, p_3^0 \ldots p_N^0\}$ and $Q^0 = \{q_1^0, q_2^0, q_3^0 \ldots q_N^0\}$

**While** termination criteria $\neq$ True:

Calculate the rank for each individual

Select N number of best individuals from $P^0$ and $Q^0$, denoted as $P^1$

Generate $Q^1$ from $P^1$ based on mutation and crossover operators

**Return** $N$ number of Pareto Fronts

Determine the most ideal solution by TOPSIS

**Return Output**

**END**

**Fig. 4.** Pseudocode for IBKA-NGBoost-NSGA-III.

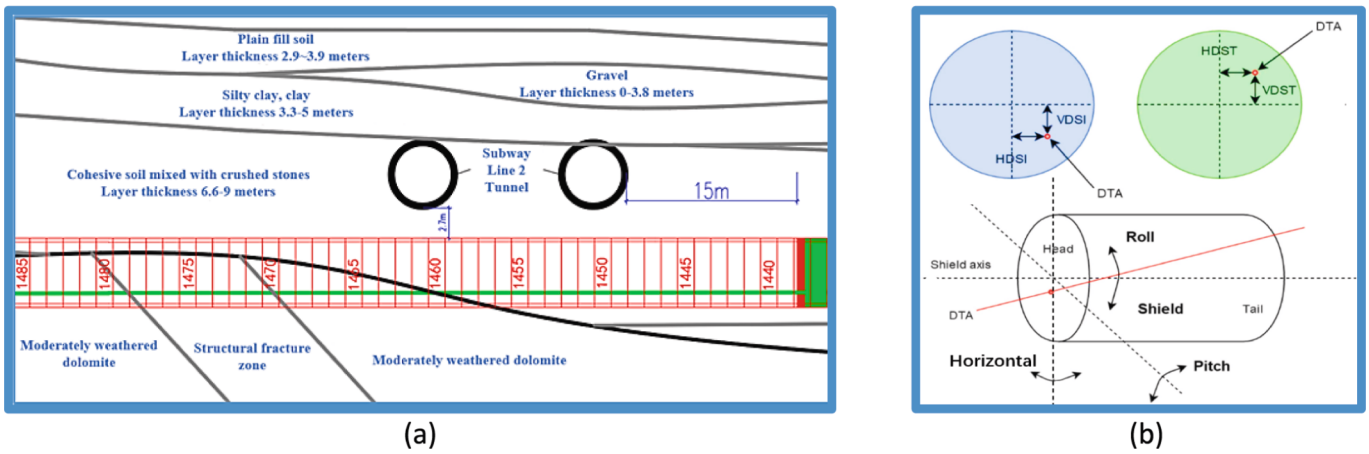



**Fig. 5.** Project background introduction: (a) Soil distribution and (b) shield machine attitude parameters.

used to calculate the geological parameters can be found in the literature [7]. Table 1 displays the data for each variable.

### 3.2.2. *Data preprocessing*

Noise reduction for construction data and shield attitude data is essential for improving prediction model performance. An analysis of the dataset reveals that data for multiple individual parameters that correspond to several ring numbers are missing. To address the effects of missing values, outliers, and noise, a series of data processing steps are implemented, including the removal of outliers, the imputation of missing values, and the application of denoising methods. Box plots are then used for processing. Taking several input parameters and 7 shield tunnel pose parameters as examples, box plots are used for detection, and the results are shown in Figs. 6 and 7.

After outliers are detected, denoising techniques are employed on the anomalous data to decrease the detrimental effects of these outliers on the model's predictive performance. In this study, the WT is used to denoise the original data, and the signal comparison results for the shield tunnel attitude parameters prior to and following denoising are shown in Fig. 8. In addition, the original shield machine position and attitude data have sharp edges and noise (Fig. 8). After threshold denoising on the basis of wavelet analysis, the obtained shield machine attitude signal is smoother, and the smoothness of the data curve is greatly improved compared with that before denoising.

After raw data cleaning and outlier removal, a Min-Max normalization procedure is applied to scale all variables into a consistent interval, thereby enhancing the numerical stability and convergence speed of the models. To rigorously evaluate predictive performance in a real-world engineering context, the dataset is partitioned into a training set and a testing set at a ratio of 8:2. Crucially, this partition follows a strict chronological sequence—training the model with historical excavation cycles to predict the forthcoming attitude parameters—rather than employing random shuffling. This time-series splitting strategy ensures that the model's generalization capability is tested considering the dynamic evolution of geological conditions. To maintain absolute evaluation fairness, all baseline models (e.g., BPNN, RF, and LSTM) are

**Table 1**
Statistical characteristics of the sample data.

| Parameter type | Variable name | Unit | Max. | Min. | Average value | Standard deviation |
|---|---|---|---|---|---|---|
| Input variables | Soil warehouse pressure—upper, X1 | bar | 2.9 | 0.1 | 1.35 | 0.33 |
| | Soil warehouse pressure—lower, X2 | bar | 6.3 | 0 | 2.19 | 1.19 |
| | Soil warehouse pressure—left, X3 | bar | 23.1 | 0.2 | 1.51 | 0.82 |
| | Soil warehouse pressure—right, X4 | bar | 3.2 | 0.1 | 1.50 | 0.47 |
| | Thrust difference between the upper and lower jacks, X5 | kN | 244 | −203 | 41.19 | 73.68 |
| | Left and right jack thrust difference, X6 | kN | 145 | −229 | −27.75 | 67.62 |
| | Travel difference between the upper and lower jacks, X7 | mm | 92 | −125 | −11.95 | 18.28 |
| | Left and right jack travel difference, X8 | mm | 82 | −108 | −1.55 | 17.77 |
| | Articulation cylinder stroke difference (horizontal), X9 | mm | 95 | 0 | 25.65 | 19.30 |
| | Articulation cylinder stroke difference (vertical), X10 | mm | 85 | 0 | 21.20 | 16.07 |
| | Articulation oil pressure, X11 | bar | 365 | 0.3 | 112.40 | 84.83 |
| | Cutterhead rotation speed, X12 | r/min | 2.9 | 0.5 | 1.65 | 0.16 |
| | Cutterhead torque, X13 | kN·m | 3200 | 650 | 2040.75 | 517.80 |
| | Advancing speed, X14 | mm/min | 80 | 3.6 | 35.69 | 14.70 |
| | Grouting pressure, X15 | bar | 3.5 | 0.7 | 2.48 | 0.37 |
| | Grouting volume, X16 | $m^3$ | 8 | 5 | 7.35 | 0.40 |
| | Spiral rotation speed, X17 | rpm | 49 | 0 | 10.22 | 5.68 |
| | Screw torque, X18 | kN·m | 201.6 | 65 | 135.96 | 27.36 |
| | Internal friction angle, X19 | ° | 30 | 36 | 33 | 2.12 |
| | Soil cohesion, X20 | kPa | 37 | 42 | 39 | 1.44 |
| | Compression modulus, X21 | MPa | 19 | 28 | 23 | 2.57 |
| Output variables | Horizontal deviation of shield machine incision, F1 | mm | 61 | −65 | −2.38 | 16.91 |
| | Vertical deviation of shield machine incision, F2 | mm | 53 | −77 | −4.55 | 27.42 |
| | Horizontal displacement of the shield machine tail, F3 | mm | 44 | −69 | −26.92 | 22.79 |
| | Vertical displacement of the shield machine tail, F4 | mm | 78 | −49 | 17.46 | 21.93 |
| | Pitch attitude angle, F5 | mm/m | 24.8 | −28 | −1.61 | 13.15 |
| | Horizontal attitude angle, F6 | mm/m | 21.6 | −15 | −1.10 | 3.82 |
| | Rolling attitude angle, F7 | mm/m | 16 | −15 | −0.03 | 3.57 |

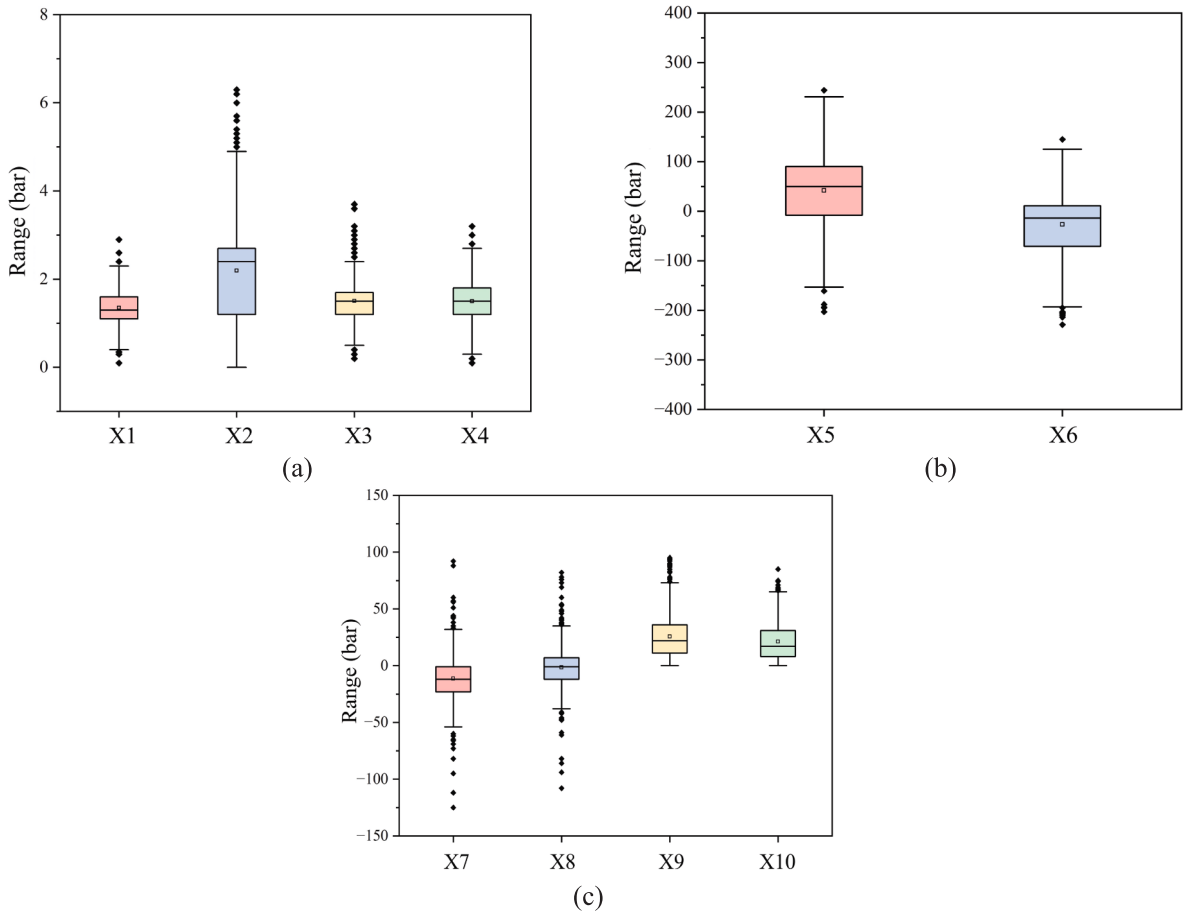



**Fig. 6.** Box plots of the input parameters (X1 ~ X10): (a) X1 ~ X4, (b) X5 ~ X6, and (c) X7 ~ X10.

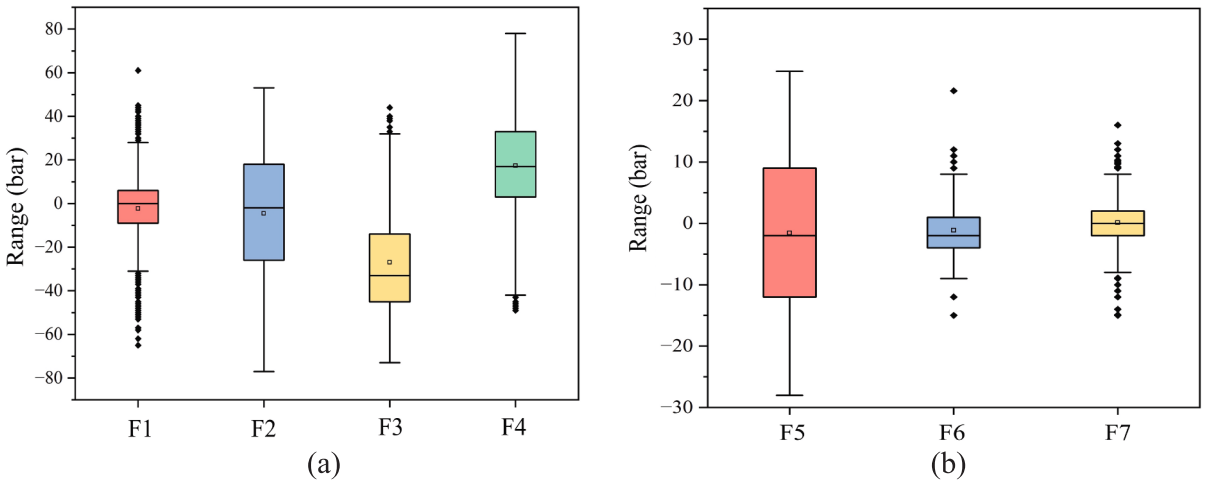



**Fig. 7.** Box plots of the shield attitude (F1 ~ F7): (a) F1 ~ F4 and (b) F5 ~ F7.

implemented using this same method of data splitting and synchronization of preprocessing parameters. The finalized training model is then deployed to forecast the multidimensional shield attitude of the test samples, facilitating a direct comparison with the high-precision field measurement data.

### 3.2.3. *Data correlation analysis*

Pearson correlation analysis reveals no significant linear associations between the construction parameters and the attitude parameters of the TBM. A heatmap that illustrates the correlation analysis results for the attitude parameters is shown in Fig. 9. According to Fig. 9, absolute values of less than 0.6 indicate that the linear correlations between the input variables are weak and relatively independent. The highest negative correlation between the advancing speed and the PAA is −0.55, and the highest positive correlation between the screw machine speed and the VDST is 0.44.

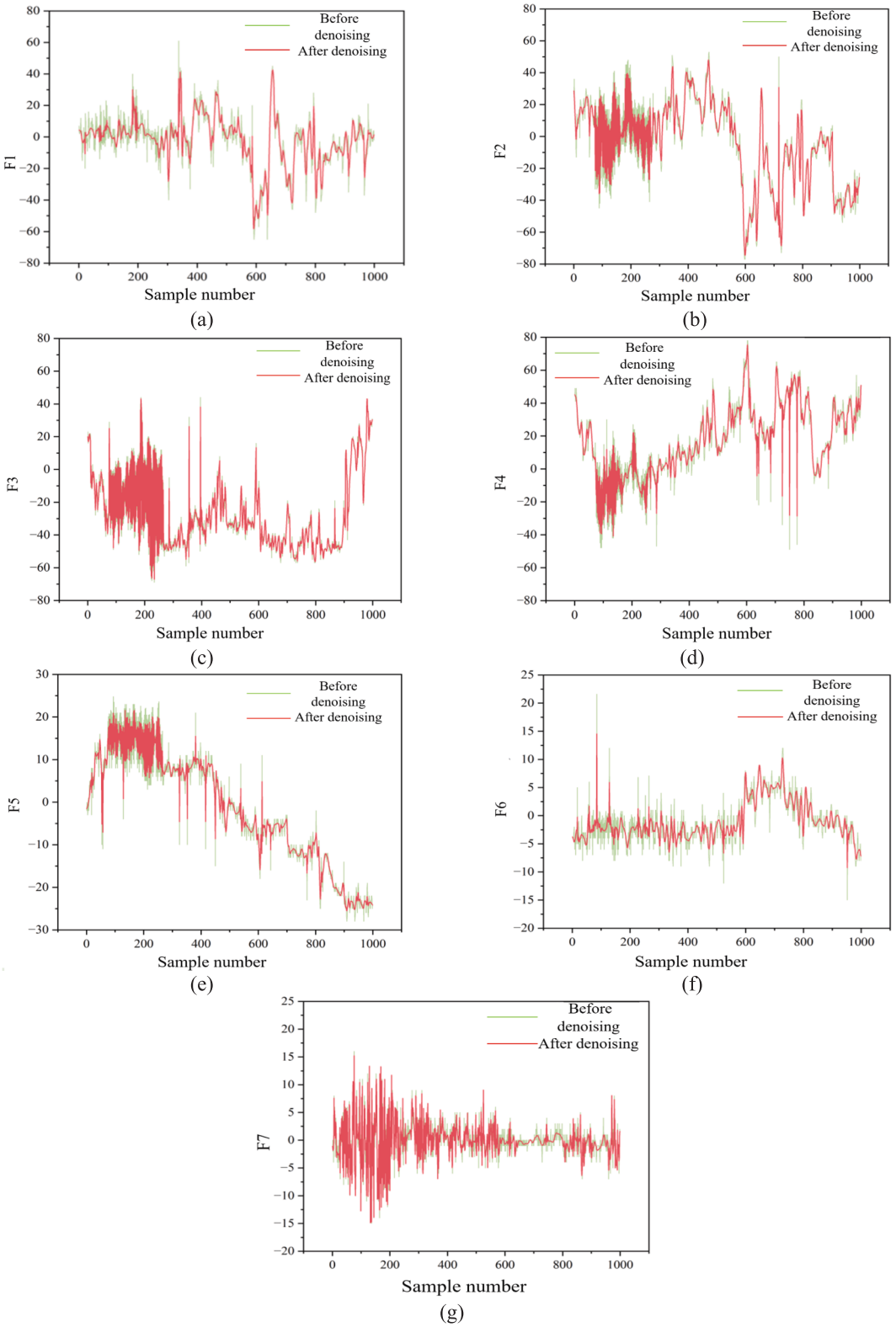



**Fig. 8.** Comparison of signals before and after denoising of the shield tunnel attitude parameters: (a) F1, (b) F2, (c) F3, (d) F4, (e) F5, (f) F6, and (g) F7.

### *3.3. Prediction of the shield attitude via IBKA-NGBoost*

#### *3.3.1. Hyperparameter optimization*

The setting of hyperparameters significantly affects model learning and training effectiveness [73,74]. The key hyperparameters of NGBoost, including learning_rate, minibatch_frac, col_sample, and n_estimators, are optimized with the IBKA. Table 2 presents the results of hyperparameter optimization for the NGBoost model. During parameter tuning, IBKA refines the parameter sampling strategy on the basis of previous model evaluation results, such as loss values, to progressively determine the optimal parameters. This iterative adjustment of search space is illustrated in Fig. 10.

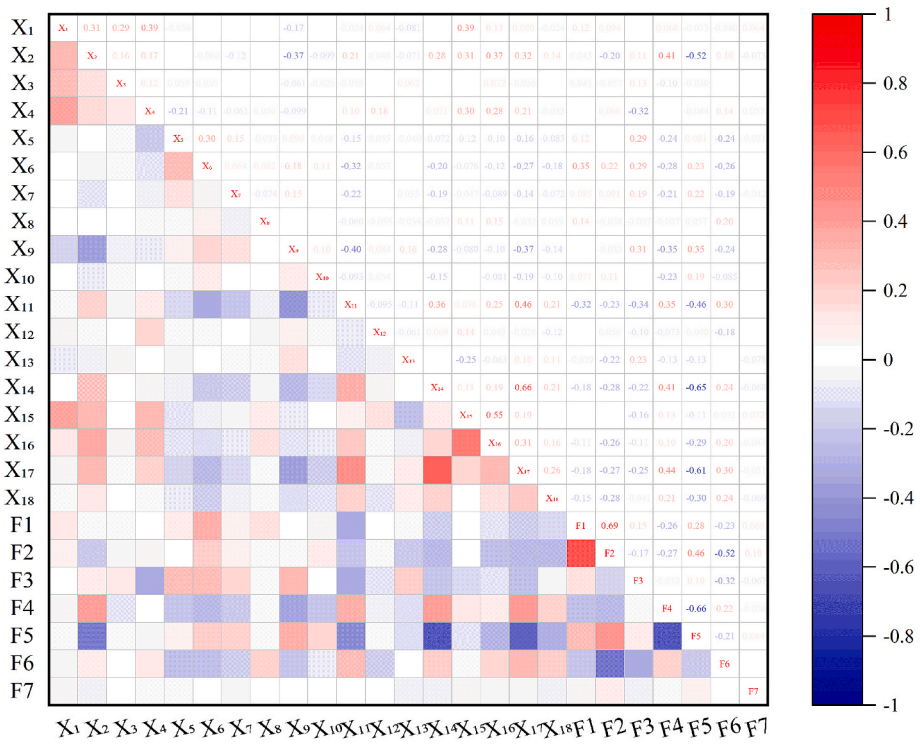



**Fig. 9.** Heatmap of the Pearson correlation coefficients between the input and output variables.

**Table 2**
Hyperparameter range settings and optimization results for NGBoost.

| Prediction target | Hyperparameters (optimization range) | | | | MSE | |
|---|---|---|---|---|---|---|
| | Learning_rate | Minibatch_frac | Col_sample | N_estimators | | Iterations |
| | (0,0.05) | (0.4,0.8) | (0.4,0.8) | (80, 200) | | |
| HDSI (F1) | 0.0397 | 0.7304 | 0.7297 | 200 | 23.94 | 89 |
| VDSI (F2) | 0.0391 | 0.7136 | 0.4952 | 200 | 82.60 | 29 |
| HDST (F3) | 0.0354 | 0.6926 | 0.5369 | 200 | 34.31 | 57 |
| VDST (F4) | 0.0371 | 0.4573 | 0.4393 | 200 | 32.95 | 23 |
| PAA (F5) | 0.0396 | 0.6123 | 0.7903 | 200 | 9.92 | 43 |
| HAA (F6) | 0.0398 | 0.5319 | 0.7305 | 200 | 0.94 | 35 |
| RAA (F7) | 0.0399 | 0.6761 | 0.4851 | 200 | 1.55 | 67 |

IBKA performs efficient and stable optimization of the hyperparameters and identifies the best combination of hyperparameters for the NGBoost model within 100 iterations. The data in Table 2 and Fig. 10 show that in the hyperparameter optimization process for the 7 objectives, the maximum number of iterations is 89. IBKA can automatically find the optimal hyperparameter combination in a small number of iterations, thus achieving high precision and high efficiency in the intelligent parameter adjustment process.

### 3.3.2. *Analysis of the prediction results*

By using the optimization outcomes that are obtained from the IBKA process, the training set is employed to train IBKA-NGBoost, which is subsequently validated using the test set to assess its performance. However, performing settlement prediction via ML models solely on the training dataset can lead to biased evaluations. Therefore, we employ k-fold cross-validation (CV) to verify the predictive results of our models, which helps ensure that the models generalize well and provides a more robust evaluation by splitting the dataset into $k$ subsets or folds and iteratively training and validating the model on different folds. A crucial parameter in k-fold CV is the number of folds k. On the basis of previous research and the distribution of our dataset, we choose $k = 5$ for this study. Cross-validation significantly increases the reliability and accuracy of our model's predictive performance. Scatter plots of the 7 target prediction results are displayed in Fig. 8. Table 3 and Fig. 11 summarize the fit quality that is indicated by $R^2$, as well as the RMSE and MAE. These metrics reflect the precision and error rates between the training and testing sets.

The values that are estimated by the NGBoost model for the 7 shield tunnel pose parameters on both the training and testing sets are highly similar to the actual values, which indicates strong model performance. The $R^2$ values for the training and testing sets range from 0.915 ~ 0.970, the RMSE range for the training and testing sets is 0.057 ~ 0.585, and the MAE range for the training and testing sets is 0.042 ~ 1.413. The data that are presented here show that the error indicators for the HDSI, VDSI, HDST, VDST, PAA, RAA, and HAA are all very low, whereas the robustness indicators are all very high. These findings indicate that the developed prediction model can yield dependable and precise results for forecasting the attitude of shield tunneling.

### 3.3.3. *Importance ranking*

The traditional correlation coefficient analysis method has the limitation of reflecting only linear relationships between features, whereas ML models can represent the complex nonlinear relationships among features. An advantage of applying gradient boosting algorithms for

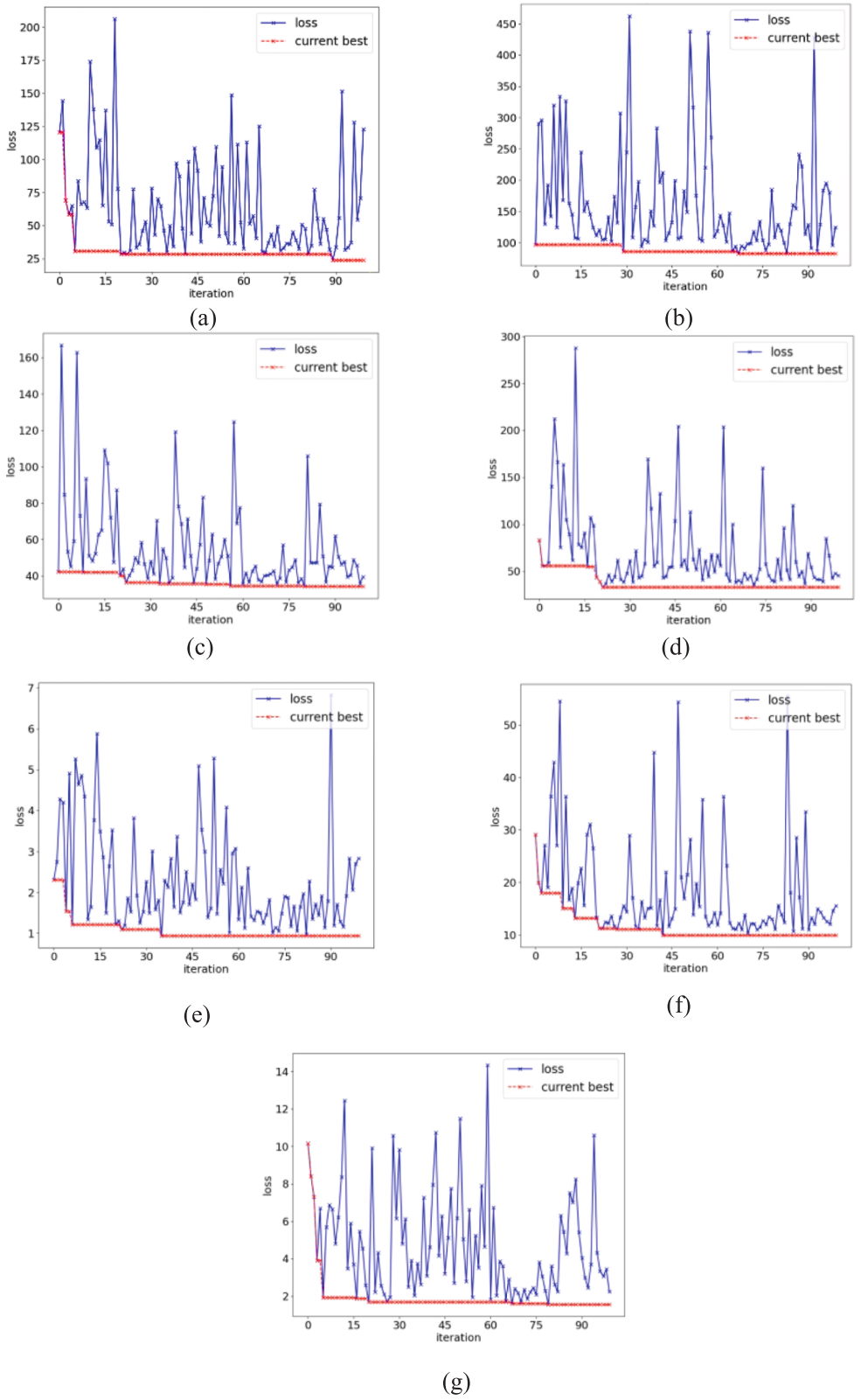



**Fig. 10.** MSE variation during the hyperparameter optimization process: (a) F1, (b) F2, (c) F3, (d) F4, (e) F5, (f) F6, and (g) F7.

prediction lies in the ability of the algorithms to use the created boosting trees to automatically provide a quantitative assessment of feature importance for the trained prediction model.

In this section, an automatic feature importance ranking algorithm is introduced to rank the influencing factors in terms of significance. First, the 21 parameter features to be evaluated are input into the NGBoost model. These TBM parameters are subsequently used as target values to train the tree models. NGBoost is employed to construct a process model for generating importance scores for various targets, and the sorted importance scores of the influencing factors are output. This approach

**Table 3**
Predictive performance of IBKA-NGBoost on the datasets.

| Dataset type Output parameters /Accuracy indicators | | Training set | Test set |
|---|---|---|---|
| HDSI (F1) | $R^2$ | 0.955 | 0.923 |
| | RMSE | 0.071 | 0.521 |
| | MAE | 0.051 | 0.399 |
| VDSI (F2) | $R^2$ | 0.943 | 0.915 |
| | RMSE | 0.153 | 0.744 |
| | MAE | 0.130 | 0.585 |
| HDST (F3) | $R^2$ | 0.970 | 0.938 |
| | RMSE | 0.111 | 0.870 |
| | MAE | 0.089 | 0.698 |
| VDST (F4) | $R^2$ | 0.962 | 0.941 |
| | RMSE | 0.105 | 0.653 |
| | MAE | 0.082 | 0.521 |
| PAA (F5) | $R^2$ | 0.963 | 0.933 |
| | RMSE | 0.097 | 0.687 |
| | MAE | 0.075 | 0.535 |
| HAA (F6) | $R^2$ | 0.941 | 0.929 |
| | RMSE | 0.060 | 0.341 |
| | MAE | 0.048 | 0.236 |
| RAA (F7) | $R^2$ | 0.953 | 0.917 |
| | RMSE | 0.057 | 0.430 |
| | MAE | 0.042 | 0.329 |

enables an understanding of the relative significance of each TBM parameter in predicting target variables such as the HDSI, VDSI, HDST, VDST, PAA, HAA, and RAA.

Using F1 ~ F7 as the prediction indices, the shield tunneling construction parameters that significantly affect the four position parameters (HDSI, VDSI, HDST, and VDST) and the three attitude parameters (PAA, HAA, and RAA) are identified as the main optimization parameters. These parameters are used to propose an executable construction plan to optimize shield tunneling operations. The findings regarding the significance of the seven shield tunnel attitude indicators are depicted in Fig. 12. This figure depicts the relative significance of the selected input factors in the context of shield tunnel performance and attitude control. The conclusions are as follows:

(1) Fig. 12 shows that the importance levels of the 21 input parameters that are selected in this study for the 7 target variables, namely, the HDSI, VDSI, HDST, VDST, PAA, HAA, and RAA, are not completely the same. However, there is a common pattern in the distribution of factors with higher importance for different attitude indicators. On the basis of the importance measurement results shown in Fig. 12**(a) to (g)**, factors such as the soil pressure, difference in jack thrust, advancing speed, cutterhead torque, and difference in jack stroke have significant effects. For example, regarding the positional deviation of shield tunneling machines, in Fig. 12(a), the importance scores of six parameters, namely, the difference in thrust between the left and right jacks, the difference in stroke between the left and right jacks, the pressure in the soil chamber—left, the pressure in the soil chamber—right, the pushing speed, and the torque of the cutterhead, are relatively large; thus, they significantly affect the HDSI.
(2) The difference in thrust between the upper and lower jacks, the pressure in the upper soil chamber, the pressure in the lower soil chamber, and the difference in stroke between the jacks, and the advancing speed are among the top parameters in the importance ranking for the vertical deviation index of the shield machine incision. With respect to the angle deviation of the TBM, Fig. 12 (e) shows that six parameters, namely, the soil chamber pressure—lower, the soil chamber pressure—upper, the cutterhead torque, the travel disparity between the upper and lower jacks, the thrust difference between the upper and lower jacks, and the advancing speed, have relatively large effects on the PAA. In Fig. 12(f), the difference in thrust between the left and right jacks, the soil chamber pressure—left, the soil chamber pressure—right, the travel disparity between the left and right jacks, and the soil chamber pressure—upper are shown. Six parameters have a significant effect on HAA. In addition, among the selected shield machine construction parameters, the importance of parameters such as articulated oil pressure, screw machine torque, and articulated deviation (horizontal and vertical) for all the shield machine attitude indicators is generally relatively small.
(3) Following the above analysis, the parameters that significantly influence the importance of the shield attitude indicators are the soil pressure, difference in jack thrust, advancing speed, cutterhead torque, and difference in jack travel. The optimization goal is to minimize the absolute values of the seven shield attitude parameters. Scenario analysis and optimization of the shield attitude are carried out to better control and adjust the construction parameters, which is conducive to maintaining the stable attitude of the shield tunneling machine.

### 3.4. Online adaptive optimization of the shield attitude

Real-time information on the operational status of the TBM is recorded and extracted through the DT platform. On the basis of prediction, intelligent optimization is carried out. Like the prediction aspect, the optimization aspect still needs to be achieved through ML and DT. The intelligent optimization module is integrated with the DT model to evaluate the key performance indicators of the TBM precisely and conduct adaptive optimization control of the shield attitude. The intelligent module includes two functions: TBM performance assessment and attitude optimization. Intelligent optimization is performed to derive initial Pareto solutions for each scenario utilizing the developed IBKA-NGBoost-NSGA-III and to acquire the optimization intervals. Afterward, the TOPSIS method is employed to facilitate optimal decision-making, and the optimization outcomes are presented on the DT platform to provide a foundation for intelligent decision-making.

#### 3.4.1. Intelligent optimization on the basis of IBKA-NGBoost-NSGA-III

1. Design of the optimization scheme

Selecting and optimizing critical excavation parameters that are adjustable in practice and strongly correlated with the attitude of the shield machine can facilitate more efficient and accurate control of deviations in this attitude. The results of an importance analysis indicate that cutterhead torque significantly affects attitude indicators. However, this parameter is inherently determined by the pressurized cutterhead of the TBM and lacks active control mechanisms. Taking this into consideration, we select 12 important active control parameters for the shield tunnel position and attitude, namely, the jack thrust (X5 ~ X6), jack stroke (X7 ~ X8), soil pressure (X1 ~ X4), cutterhead speed (X12), advancing speed (X14), and grouting (X15 ~ X16), for adjustment, which results in 5 scenarios, as shown in Table 4.

2. Determining the objective function

The relevant regulations require that the maximum discrepancy between the axis of the shield machine and the design axis of the tunnel during shield tunneling be ± 50 mm. In practical engineering, the directional discrepancy of the shield machine must generally be maintained at or below 30 mm to ensure that the tunnel position is within the allowable range. The allowable deviation limit for the PAA and HAA is ± 20 mm/m, and the rolling angle is limited to within 10 mm/m. The deviations of some of the shield tunnel attitude parameters exceed the limit ranges, and the axis of the shield machine deviates far from the design axis. Correcting and adjusting the shield tunnel attitude is necessary, and there is a large optimization space.

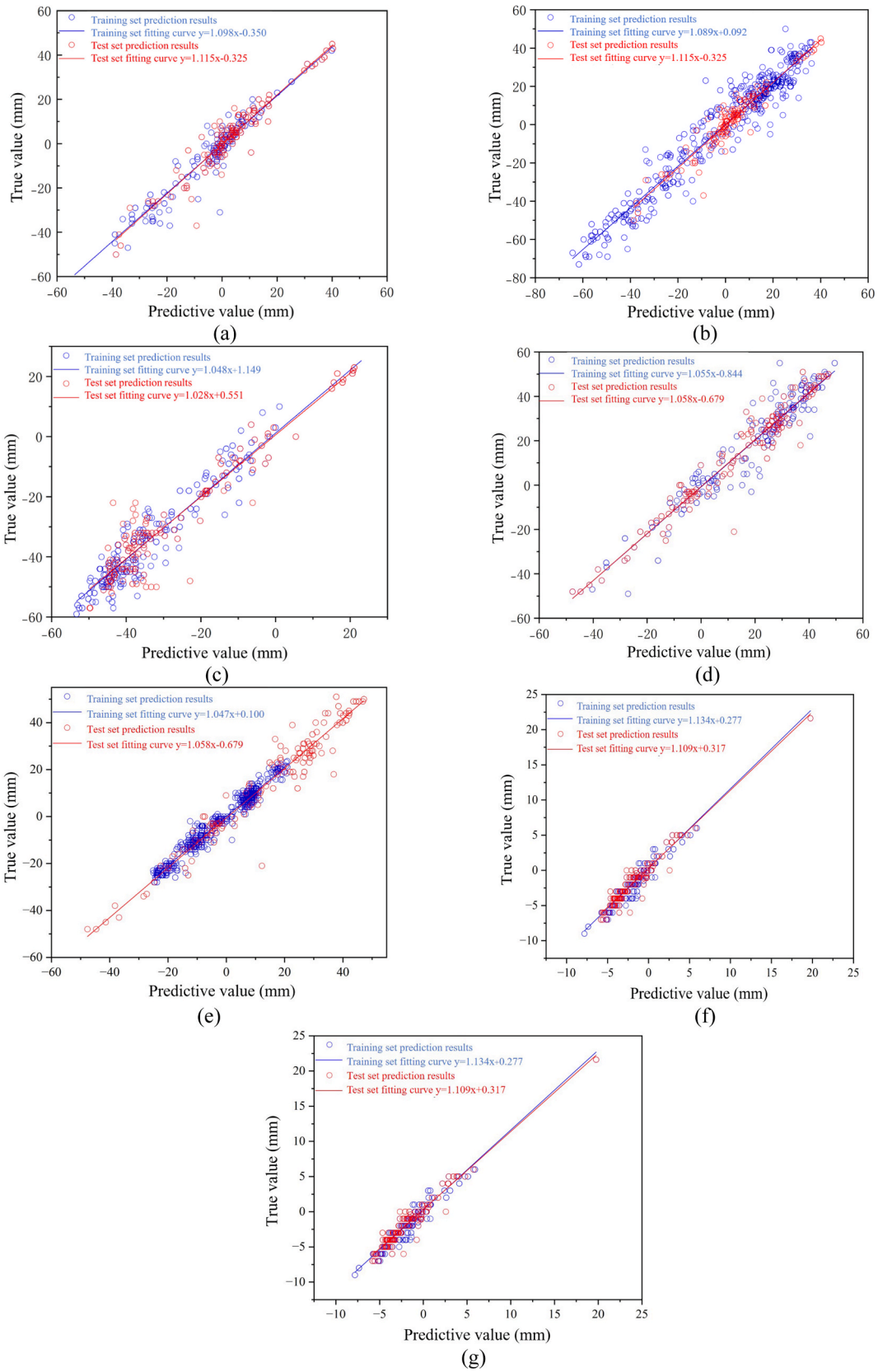



**Fig. 11.** Prediction results for the shield machine pose parameters: (a) F1, (b) F2, (c) F3, (d) F4, (e) F5, (f) F6, and (g) F7.

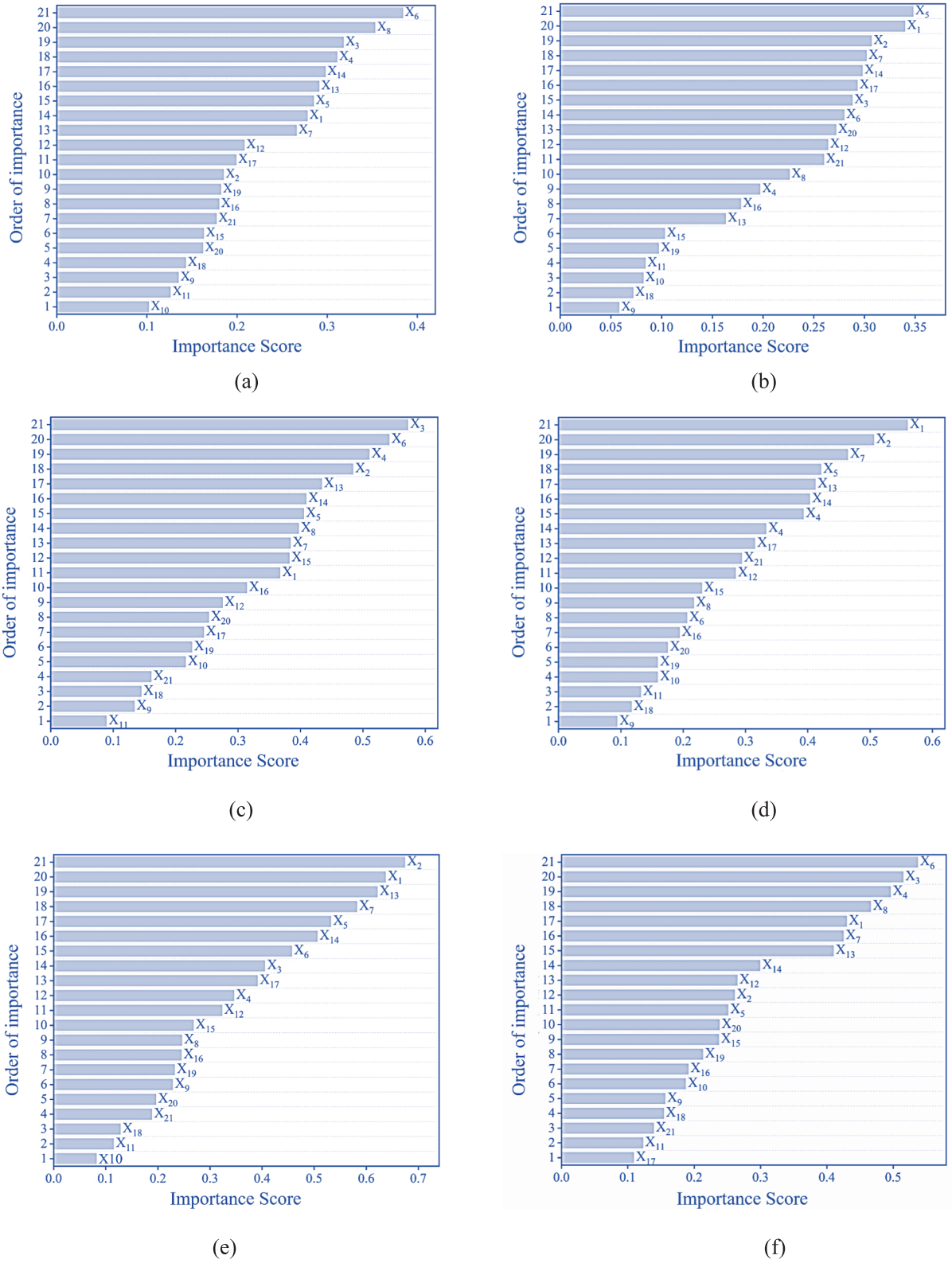



**Fig. 12.** Importance analysis of influencing factors of the shield tunnel pose parameters: (a) HDSI, (b) VDSI, (c) HDST, (d) VDST, (e) PAA, (f) HAA, and (g) RAA.

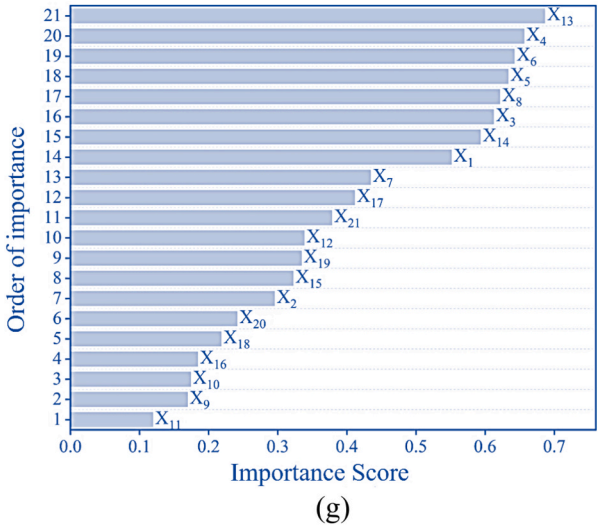



**Fig. 12.** (*continued*).

**Table 4**
Optimization strategies for the five scenarios.

| Scenario | Adjustable parameters | Scenario | Adjustable parameters |
|---|---|---|---|
| Scenario 1 | X5 ~ X6, X7 ~ X8 | Scenario 4 | X1 ~ X4, X5 ~ X6, X7 ~ X8, X12, X14 |
| Scenario 2 | X1 ~ X4, X5 ~ X6, X7 ~ X8 | Scenario 5 | X1 ~ X4, X5 ~ X6, X7 ~ X8, X12, X14, X15 ~ X16 |
| Scenario 3 | X1 ~ X4, X5 ~ X6, X7 ~ X8, X12 | – | – |

The HDSI, VDSI, HDST, VDST, PAA, RAA, and HAA prediction functions that are acquired through the IBKA-NGBoost prediction model can serve as fitness functions for MOO. The optimization design is carried out with 7 shield tunnel pose parameter indicators as the objectives. For example, for Scenario 5, the objective function is as follows [75]:

$$\begin{aligned} &minF(X) = [F1(X), F2(X), ..., F7(X)] \\ &X_i^l \le X_i \le X_i^u \end{aligned} \tag{26}$$

$$F_i(X) = \text{IBKA} - NGBoost(X_1 \sim X_4, X_5 \sim X_6, X_7 \sim X_8, X_{12}, X_{14}, X_{15} \sim X_{16}) \tag{27}$$

where $F_i(X)$ represents the shield attitude parameters, $X_i^l$ represents the lowest assigned value for the corresponding input parameter, $X_i^u$ represents the highest designated value of the corresponding input parameter, $X_1 \sim X_4$ represent the pressure of the soil silo, $X_5 \sim X_6$ represent the jack thrust, $X_7 \sim X_8$ represent the jack stroke, $X_{12}$ represents the speed of the cutterhead, $X_{14}$ represents the advancing speed, and $X_{15}$ and $X_{16}$ represent the injection volume and injection pressure, respectively.

## 3. Determining the scope of the constraints

To ensure that the shield tunneling parameter combinations are reasonable and feasible, defining explicit constraints for each decision variable in consideration of real engineering circumstances and relevant specifications is crucial. These constraints not only reflect the physical limits of shield tunneling machine operation but also incorporate various considerations, such as construction experience, geological conditions, and equipment performance. By setting reasonable constraints, we can ensure that the optimized parameter combinations can meet the construction requirements while maintaining the secure and stable operation of the shield tunneling machine. The specific input parameter constraints are shown in Table 5, which details the value range of each parameter and the respective basis, which provides a solid foundation for subsequent parameter optimization work.

**Table 5**
Ranges of the input parameters.

| Input parameter | DT platform abbreviation | Unit | Parameter range |
|---|---|---|---|
| Soil warehouse pressure-upper (X1) | PU | bar | [4] |
| Soil warehouse pressure-lower (X2) | FLO | bar | [7] |
| Soil warehouse pressure-left (X3) | PLE | bar | [7] |
| Soil warehouse pressure-right (X4) | PR | bar | [4] |
| Thrust difference between the upper and lower jacks (X5) | THUL | kN | [-250,250] |
| Thrust difference between the left and right jacks (X6) | THLR | kN | [-250,250] |
| Travel difference between the upper and lower jacks (X7) | TRUL | mm | [-120,100] |
| Travel difference between the left and right jacks (X8) | TRLR | mm | [-120,100] |
| Cutterhead rotation speed (X12) | CRS | r/min | [3] |
| Advancing speed (X14) | AS | mm/min | [76] |
| Grouting pressure (X15) | GP | bar | [4] |
| Grouting volume (X16) | GV | m3 | [8] |

### 3.4.2. Analysis of the scenario optimization results

Drawing on the established IBKA-NGBoost-NSGA-III algorithm, the attitude during the tunnel excavation process is minimized and globally optimized. On the basis of the literature and following extensive experimentation, the crossover probability (0.7), mutation probability (0.01), population size, maximum number of generations of evolution and upper limit of the number of generations before NSGA-III is terminated (100) are set [7,70].

After the parameters are set, the IBKA-NGBoost-NSGA-III algorithm is used to generate Pareto solution sets across various optimization scenarios. The average computation time for one NSGA-III optimization run is 4.2 min on a CPU. TOPSIS is then employed to identify the best compromise solution from the Pareto frontier. In this study, all the objectives are regarded as equally important and handled within a preference-free framework, which fundamentally differs from conventional multiobjective methods that rely on subjective weight assignment. On the basis of the real-time distribution of the Pareto front, the

**Table 6**
Pareto solution for optimization in Scenario 1.

| Parameter number | X5 kN | X6 kN | X7 mm | X8 mm | F1 mm | F2 mm | F3 mm | F4 mm | F5 mm/m | F6 mm/m | F7 mm/m |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 68 | 10 | 67 | −19 | −1.8 | 0.5 | −18.7 | 17.2 | −1.4 | −0.6 | −0.5 |
| 2 | 6 | 4 | −105 | 9 | 2.2 | −1.1 | −39.7 | −1.6 | 9.2 | 0.0 | 0.1 |
| 3 | 223 | −107 | 26 | −45 | −0.5 | −21.7 | −10.4 | 0.2 | 0.1 | −1.0 | 4.9 |
| 4 | 6 | 3 | −33 | 9 | 1.8 | −3.8 | −22.4 | 3.3 | 6.7 | 0.8 | 0.6 |
| 5 | 146 | −106 | 47 | −41 | 0.3 | −2.6 | −12.3 | −0.1 | 8.9 | −4.0 | 8.6 |
| … | … | … | … | … | … | … | … | … | … | … | … |
| 50 | 122 | 3 | −9 | −19 | −1.0 | −4.4 | −0.4 | 16.5 | 0.3 | −1.5 | −0.5 |
| 51 | 182 | −102 | 56 | −47 | −4.4 | −10.7 | −9.1 | 5.0 | 3.1 | −3.5 | 4.8 |
| 52 | 244 | 37 | −31 | 9 | 4.5 | −11.3 | −38.6 | −0.4 | 5.5 | 1.6 | 3.4 |
| 53 | 105 | 4 | −98 | −18 | −18.5 | −5.8 | −11.5 | 14.0 | 3.6 | −0.8 | 0.1 |
| 54 | 146 | −106 | 39 | −41 | 0.3 | −2.6 | −12.3 | −0.1 | 8.9 | −4.0 | 8.6 |
| 55 | 161 | 0 | −111 | −19 | −22.6 | −5.8 | −25.3 | 2.6 | 4.2 | −2.3 | 0.9 |
| 56 | 94 | 2 | −35 | 11 | −3.2 | −8.2 | −3.9 | 1.5 | 3.4 | −0.1 | −0.2 |

TOPSIS approach normalizes the decision matrix to eliminate scale differences and thus selects the optimal optimization decision. This method can achieve optimization of the five scenarios of this study, which are presented in Table 4. Using Scenario 1 as a case study, during the optimization process, 56 sets of optimization solutions are obtained through an optimization search. The optimized data for each shield tunnel attitude indicator parameter are shown in Table 6. A comparison before and after attitude optimization in Scenario 1 is displayed in Fig. 13. The improvements in the 7 optimization goals in various scenarios are displayed in Table 7 and Fig. 14. A comparison of the improvement effects of each scenario is shown in Fig. 15. The results can be summarized as follows:

(1) In the context of adjusting various combinations of construction parameters, the deviations of the seven shield tunnel attitude parameters can be minimized to some extent, and the overall attitude control quality can be improved by an average of approximately 69.3%. Table 7 and Fig. 14 illustrate the improvements that are achieved compared with the mean values of the original data samples. In Scenarios 1 to 5, the F1 deviation is reduced by at least 59.7% and at most 86.9%, the F2 deviation is reduced by at least 55.4% and at most 91.8%, the F3 deviation is reduced by at least 38.8% and at most 90.0%, the F4 deviation is reduced by at least 72.8% and at most 93.9%, the F5 deviation is reduced by at least 48.7% and at most 83.7%, the F6 deviation is reduced by at least 49.5% and at most 88.7%, the minimum reduction rate of F7 is 27.5%, and the maximum is 87.2%.
(2) Modifying additional parameters that are related to shield tunneling construction can lead to substantial optimization improvements, with the most significant improvements resulting from adjustments to 12 input variables. As displayed in Fig. 15, combined with the calculation outcomes in Table 7, when adjustments are made to just four construction parameters for the shield tunnel, the average percentage improvement reaches 36.5%. In Scenario 3, adjusting 9 shield tunneling construction parameters simultaneously results in an average improvement percentage of 78.6%. In Scenario 4, adjusting 10 shield tunneling construction parameters simultaneously results in an average improvement percentage of 74.9%. In Scenario 5, adjusting 12 shield tunneling construction parameters simultaneously results in an average improvement percentage of 84.8%. These results indicate that the best optimization effect can be achieved by adjusting 12 parameters.
(3) The optimized shield machine has a more stable attitude and is more conducive to construction safety. After adaptive optimization via IBKA-NGBoost-NSGA-III, the seven shield pose targets remain within the warning value range, and none exceed the range (Fig. 13). The consistent stability observed in the optimized attitude trajectory is attributed to the synergistic effect of NGBoost's uncertainty quantification and NSGA-III's reference-point steering, which effectively filters out control actions associated with high prediction variance. The fluctuation of the output attitude value significantly decreases, which indicates that shield tunneling is more stable. The effectiveness of this adaptive regulation method when applied to the underpass construction of Wuhan Metro Line 2 has been demonstrated.

### 3.4.3. *Intelligent decision-making*

*3.4.3.1. Acquisition of the optimal solution and updates for online optimization.* In the above scenario, when the TOPSIS decision is used, adjusting 12 shield construction parameters simultaneously, i.e., Scenario 5, can achieve the optimal adjustment of the shield machine attitude index. Table 8 shows the ranges of construction parameter values that correspond to the Pareto solution set that is acquired through MOO. According to Table 8, compared with the original data size, the range of optimized operation parameters is reduced, which allows for better control and adjustment of construction parameters within this range, thus making maintaining the stable attitude of the TBM easier. The current optimal solution for ring 1660 is presented in detail as an example in Table 8.

*3.4.3.2. Intelligent decision-making presentation.* The data and construction parameters in this study are adjusted using the DT platform, which is depicted in Fig. 16. The interface display area can be divided into key sections. On the left, the actual and recommended values for the main construction parameters are shown. The actual value is a real-time monitoring value, and the recommended value is determined by an intelligent algorithm after MOO. The four disks on the right display the current actual adjusted metrics of the key construction parameters, and the four graphical panels at the bottom display the monitoring adjusted metrics of the key construction parameters. On the lower-right side, a geological cross-sectional map of the relationship between the tunnel and Line 2 is provided. With the IBKA-NGBoost algorithm, the main construction parameters that affect each target are identified, thus providing a reference for adjusting important construction parameters under different target requirements.

In conclusion, the proposed DT platform, which is integrated with intelligent modules, offers dependable guidance for improved control during the operation of a TBM. The preceding cases demonstrate that by applying the suggested online optimization approach, the DT model can deliver the recommended optimal value. Moreover, the attitude of the shield machine is effectively controlled. As shown in Table 8, the shield attitude is optimized, and the recommended construction parameters are obtained via TOPSIS. As displayed in Fig. 14, by the optimization of DT, the shield attitudes of all seven targets are controlled compared with the original values. Moreover, the DT platform that is shown in Fig. 16 (b) can display important construction parameters and attitude

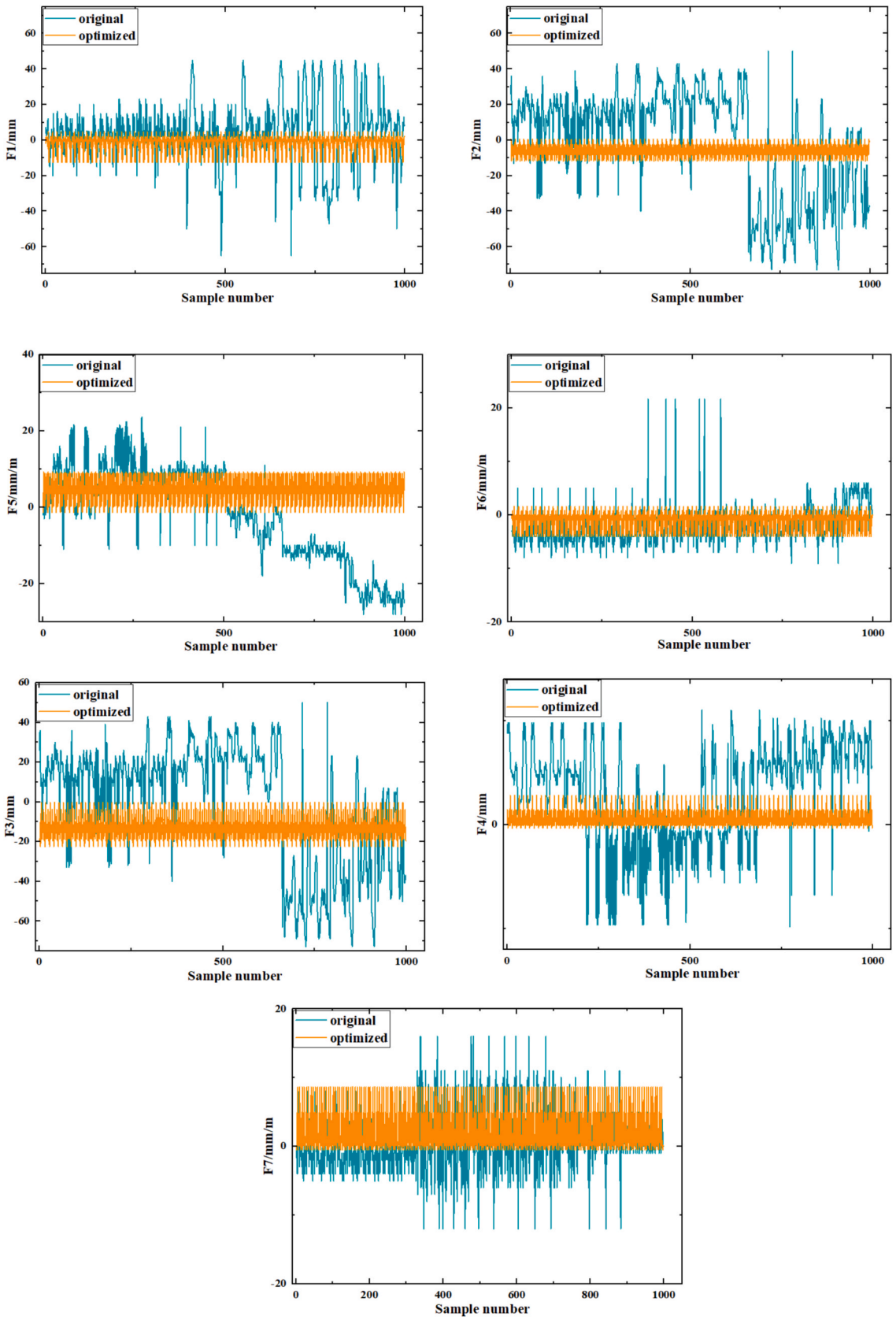



**Fig. 13.** Comparison of the attitude optimization results in Scenario 1.

**Table 7**
Improvement of the 7 Optimization Goals in Five Scenarios.

| Target | Data feature | Scenario | | | | | |
|---|---|---|---|---|---|---|---|
| | | Original | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4 | Scenario 5 |
| F1 (mm) | average value | 9.88 | 3.98 (59.7%*) | 1.66 (83.2%) | 1.43 (85.5%) | 1.48 (85.0%) | 1.29 (86.9%) |
| | standard deviation | 8.20 | 5.01 | 2.10 | 3.30 | 1.82 | 1.84 |
| | 95% CI | – | 59.7 ± 4.1% | 83.2 ± 3.9% | 85.5 ± 3.7% | 85.0 ± 3.5% | 86.9 ± 3.4% |
| F2 (mm) | average value | 18.80 | 8.39 (55.4%) | 1.97 (89.5%) | 1.55 (91.8%) | 5.22 (72.3%) | 2.31 (87.7%) |
| | standard deviation | 10.69 | 9.11 | 1.98 | 1.75 | 6.39 | 3.80 |
| | 95% CI | – | 55.4 ± 3.8% | 89.5 ± 3.6% | 91.8 ± 3.4% | 72.3 ± 3.7% | 87.7 ± 3.3% |
| F3 (mm) | average value | 25.38 | 15.54 (38.8%) | 5.80 (77.2%) | 5.04 (80.2%) | 3.69 (85.5%) | 2.53 (90.0%) |
| | standard deviation | 15.27 | 14.65 | 9.06 | 6.85 | 5.26 | 4.50 |
| | 95% CI | – | 38.8 ± 4.0% | 77.2 ± 3.8% | 80.2 ± 3.6% | 85.5 ± 3.3% | 90.0 ± 3.2% |
| F4 (mm) | average value | 20.70 | 5.62 (72.8%) | 4.44 (78.6%) | 2.67 (87.1%) | 2.68 (87.0%) | 1.27 (93.9%) |
| | standard deviation | 13.52 | 6.36 | 7.97 | 5.28 | 3.03 | 2.40 |
| | 95% CI | – | 72.8 ± 3.9% | 78.6 ± 3.7% | 87.1 ± 3.5% | 87.0 ± 3.3% | 93.9 ± 3.0% |
| F5 (mm/m) | average value | 9.58 | 4.91 (48.7%) | 1.90 (80.2%) | 1.79 (81.3%) | 1.78 (81.4%) | 1.56 (83.7%) |
| | standard deviation | 5.97 | 3.92 | 3.92 | 2.36 | 1.55 | 1.96 |
| | 95% CI | – | 48.7 ± 4.2% | 80.2 ± 3.8% | 81.3 ± 3.6% | 81.4 ± 3.5% | 83.7 ± 3.3% |
| F6 (mm/m) | average value | 3.06 | 1.54 (49.5%) | 0.54 (82.2%) | 0.35 (88.7%) | 0.42 (86.4%) | 0.36 (88.2%) |
| | standard deviation | 2.10 | 1.80 | 0.61 | 0.43 | 0.65 | 0.54 |
| | 95% CI | – | 49.5 ± 4.0% | 82.2 ± 3.7% | 88.7 ± 3.4% | 86.4 ± 3.5% | 88.2 ± 3.2% |
| F7 (mm/m) | average value | 2.78 | 1.78 (36.1%) | 2.02 (27.5%) | 0.35 (87.2%) | 1.35 (51.6%) | 0.52 81.2%) |
| | standard deviation | 2.60 | 2.09 | 3.32 | 0.52 | 2.70 | 0.74 |
| | 95% CI | – | 36.1 ± 4.3% | 27.5 ± 4.5% | 87.2 ± 3.3% | 51.6 ± 4.0% | 81.2 ± 3.4% |
| Average optimization | | – | 36.5% | 71.5% | 78.6% | 74.9% | 84.8% |

parameters in real time, thus establishing a foundation for subsequent online updating. Future research should seek to improve intelligent shield tunneling by incorporating more varied data and to achieve advanced adaptive optimization of shield attitudes over a wider range of geological conditions to improve adaptation, which is explored in the discussion section of this study.

## 4. Discussion

This section provides an in-depth examination of the proposed method, with the aim of validating its effectiveness and generalizability.

### 4.1. Comparison of the predictive precision and effectiveness

To rigorously demonstrate the superior predictive ability of the IBKA-NGBoost model, a comparative analysis is conducted with seven prevalent artificial intelligence algorithms, IBKA-BPNN, IBKA-XGBoost, IBKA-RF, IBKA-LSTM, IBKA-GCN-LSTM, IBKA-GRU and NGBoost, all of which are benchmarked against IBKA-NGBoost [76–79]. These algorithms are trained to predict the shield attitude using the same dataset. The hyperparameter settings are critical to the performance of the models; thus, we select and set the optimal hyperparameters for each prediction model. The hyperparameter settings follow the descriptions and selections in past papers and are slightly modified after the experiments to adapt to our dataset. Table 9 presents the selected hyperparameters and their optimal values for each prediction algorithm, and the specific details of the hyperparameters for the IBKA-NGBoost algorithm are displayed in Table 2. The performance metrics for these eight AI algorithms in predicting seven attitude parameters of the shield are shown in Fig. 17. Table 10 presents the average durations that are needed to train and validate the eight models chosen for this study on the same device. The specific findings are as follows:

(1) Optimization via the IBKA can significantly strengthen the forecasting capability. The performance of the algorithm with IBKA optimization is better than that of the algorithm without hyperparameter optimization, which generally aligns with the conclusions that were drawn by Xiao et al. regarding the predictions of the shield attitude, and the model's predictive performance is significantly improved after IBKA is added for hyperparameter optimization [80].
(2) On the small data samples of this study, IBKA-NGBoost achieves a balance between predictive performance and efficiency. The forecasting accuracy of the various models, as displayed in Fig. 17, emphasizes the effectiveness of the IBKA-NGBoost method in improving the prediction of the attitude parameters. Specifically, the $R^2$ values of the seven attitude indicators all exceed 0.91, and the RMSE values remain lower than 0.9. Fig. 16 additionally shows that IBKA-GCN-LSTM, IBKA-Transformer and IBKA-GRU also demonstrate good predictive performance and outperform IBKA-NGBoost in terms of several indicators. However, during the model training period, these models need more training time and more investment in time and computing resources. The training times that of various prediction models are ordered as follows: IBKA-GCN-LSTM > IBKA-GRU > IBKA-Transformer > IBKA-NGBoost > NGBoost > IBKA-BPNN > IBKA-RF > IBKA-XGBoost. Furthermore, 29.2 s fully meets the requirement of collecting data once every half minute, as stipulated in the project. Among the commonly used deep learning algorithms selected, IBKA-Transformer, which takes the least time, still requires 46.9 s. Clearly, it does not meet practical engineering requirements. In terms of efficiency, IBKA-NGBoost is far superior to the deep learning algorithms. Therefore, when performance and efficiency are comprehensively considered, IBKA-NGBoost is optimal.

In summary, through a comparative study of various algorithms, the proposed intelligent prediction model demonstrates excellent performance for shield tunnel attitude prediction and accurately establishes the complex relationships between influencing factors and shield tunnel

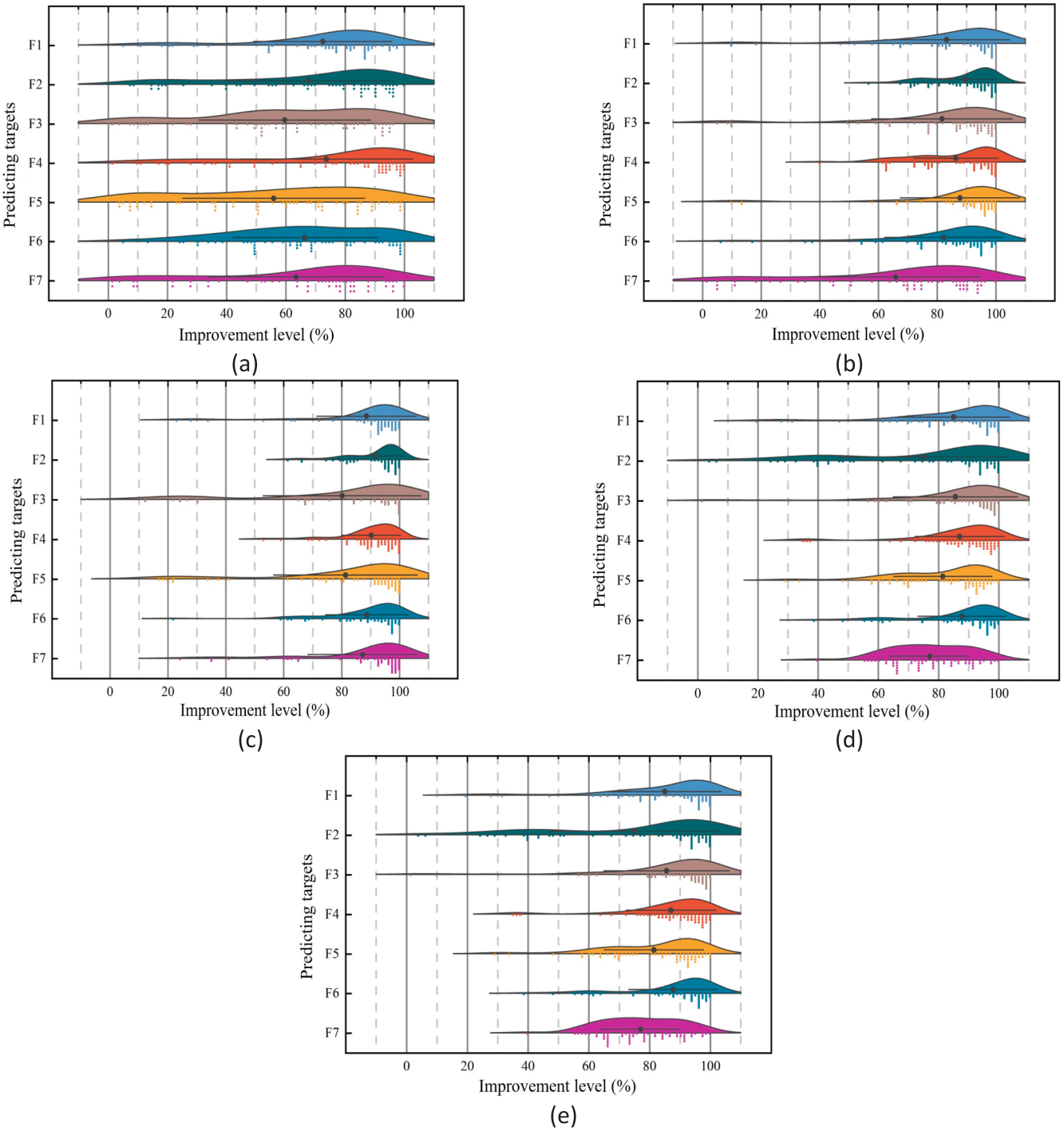



**Fig. 14.** Optimization effect of the shield attitude: (a) Scenario 1, (b) Scenario 2, (c) Scenario 3, (d) Scenario 4, and (e) Scenario 5.

attitude. Even with a restricted dataset, satisfactory prediction outcomes and high efficiency can be achieved. These findings suggest that the IBKA-NGBoost algorithm is highly suitable for simulating the relationships among shield attitude, construction factors, and soil conditions and is very effective at predicting shield attitude.

All the models were trained and tested on the same laptop: an AMD Ryzen 7 8845H processor (8 cores and 3.80 GHz), 32 GB of DDR5-5600 RAM, an integrated Radeon 780 M GPU, a Windows 11 64-bit system, and Python 3.9.13. Each reported training time is the average of five runs.

### *4.2. Generalizability of the developed hybrid intelligence algorithm*

We directly applied the original model (trained on the clay-sandstone stratum) to the same project but different sections of the quartz sandstone dataset without any retraining or fine-tuning. We conducted a rigorous cross-condition verification to test the domain transfer robustness of the model. The IBKA-NGBoost-NSGA-III model is used to analyze the geology of quartz sandstone, and prediction optimization is conducted for Scenario 1, which is similar to that in Section 3.5, across 7 targets with attitudes on the basis of 3,600 sample data points. The evaluation results for the prediction performance are displayed in Table 11, and the optimization outcomes are displayed in

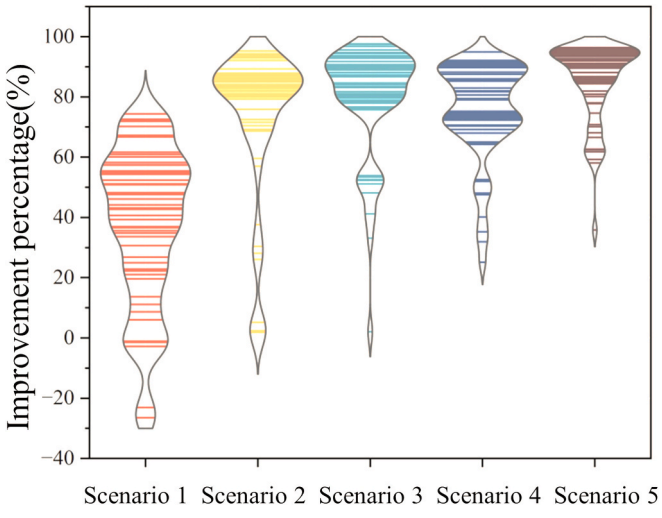



**Fig. 15.** Comparison of improvement levels for Scenarios 1–5.

**Table 8**
Value ranges of the operational parameters before and after optimization.

| Variable | Operational parameter | Original interval | Optimization interval | Current optimal solution |
|---|---|---|---|---|
| X1/bar | Soil warehouse pressure-upper (PU) | [0.1,2.9] | [1.0,2.8] | 2.0 |
| X2/bar | Soil warehouse pressure-lower (FLO) | [0,6.3] | [1.4,3.3] | 1.9 |
| X3/bar | Soil warehouse pressure-left (PLE) | [0,3.7] | [0.5,1.6] | 2.0 |
| X4/bar | Soil warehouse pressure-right (PR) | [0.1,3.2] | [0.2,1.4] | 2.0 |
| X5/kN | Left and right jack thrust difference (THUL) | [-203,244] | [-203,60] | −50 |
| X6/kN | Differential thrust between the upper and lower jacks (THLR) | [-229,145] | [-67,19] | −60 |
| X7/mm | Travel difference between the left and right jacks (TRUL) | [-125,92] | [-81,37] | 20 |
| X8/mm | Travel difference between the upper and lower jacks (TRLR) | [-108,82] | [9,79] | 30 |
| X12/r/min | Cutterhead rotation speed (CRS) | [0.5,2.9] | [0.5,1.4] | 1.2 |
| X14/mm/min | Advancing speed (AS) | [5,79] | [12,21] | 15 |
| X15/bar | Grouting pressure (GP) | [0.8,3.5] | [0.8,1.8] | 1.6 |
| X16/m3 | Grouting volume (GV) | [5,8] | [5.4,7.1] | 6.2 |

Table 12. The findings are as follows:

(1) The developed IBKA-NGBoost model has a high degree of predictive accuracy and significant applicability. In the context of a quartz sandstone environment, the developed prediction model for attitude yields $R^2$ values that range from 0.910 to 0.952, thus effectively capturing the nonlinear relationships between the attitude and input parameters. The findings suggest that the model is applicable for attitude prediction under two distinct soil conditions, thereby demonstrating broad applicability, elevated prediction accuracy and significant generalizability.
(2) IBKA-NGBoost-NSGA-III is well suited for optimizing engineering attitudes in quartz sandstone and features an optimization range of 11.9% to 48.6% across the seven poses. The developed IBKA-NGBoost-NSGA-III model is applicable for forecasting and optimizing construction parameters across two prevalent geological conditions and offers valuable insights for projects in similar soil contexts. Nonetheless, attitude control is a complex process that is influenced by various factors, including construction parameters, soil quality, moisture content, burial depth, and the skill level of construction personnel.

Grounded in a multidimensional comparative analysis, the proposed method shows considerable benefits in terms of prediction performance, computing efficiency and the generalization of adaptive optimization, thus providing a feasible basis for integration into the DT platform.

### 4.3. Performance evaluation of the constructed system

To rigorously evaluate the superiority of the proposed DT system, a controlled comparative study was conducted. We selected a tunneling section adjacent to our case study area, ensuring comparable geological conditions (i.e., a clay-sandstone stratum) and tunnel design parameters. A control group was established where tunneling operations were managed by a team of seasoned engineers relying solely on a conventional guidance system (the “manual decision-making mode”). This group operated independently without access to the DT platform's optimization suggestions. The comparison spanned a continuous 48-hour period under similar advance rates, ensuring the reliability of the comparative results. Table 13 presents the average deviations of the seven attitude targets for both groups during this period.

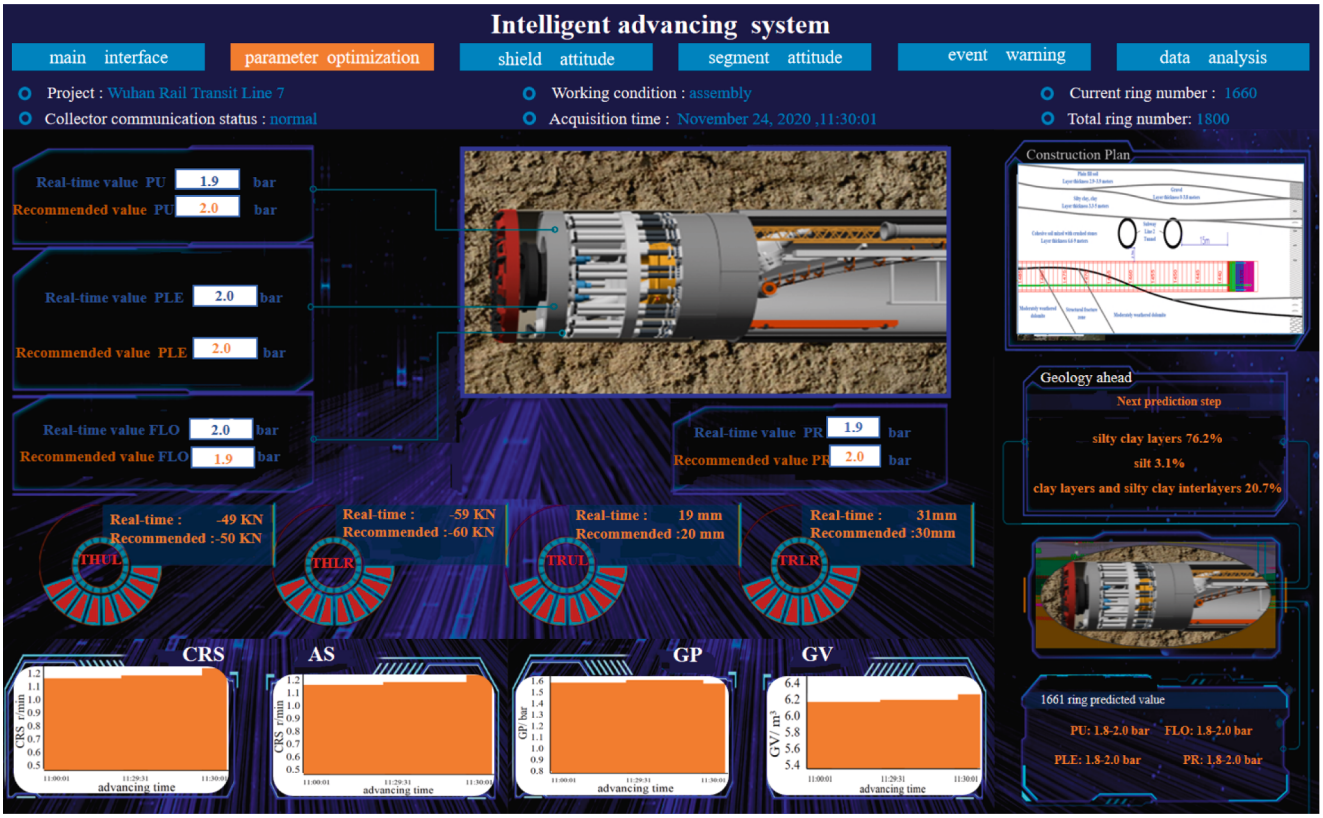
The results indicate a consistent reduction in attitude deviations across all seven targets, with improvement rates ranging from 21.10% to 28.81% (Table 13). This superior performance can be attributed to the core intelligence of the DT system. Unlike the manual mode, which often reacts to deviations after they occur, the IBKA-NGBoost model enables proactive prediction of attitude trends. Subsequently, the IBKA-NGBoost-NSGA-III algorithm performs real-time MOO, generating optimal parameter sets before significant deviations manifest. This closed-loop “perception-analysis-optimization-feedback” mechanism fundamentally narrows the “perception-action” gap, transitioning the control paradigm from passive correction to active guidance.

Beyond the quantitative metrics, this system-level evaluation underscores the DT framework's role as a functional decision-support system rather than a mere data visualization tool. The high reliability (99.5% uptime) and low latency (<5 s) confirm its robustness for real-time field applications. However, the current system's performance is inherently tied to the quality and quantity of sensor data. Future work will focus on enhancing the system's adaptability to more complex and variable geological conditions by incorporating transfer learning techniques and enriching the digital model with real-time geological prospecting data. Furthermore, exploring human-in-the-loop collaboration models, where the system provides decision options and engineers make the final call, could be a promising direction for balancing automation efficiency and operational expertise.

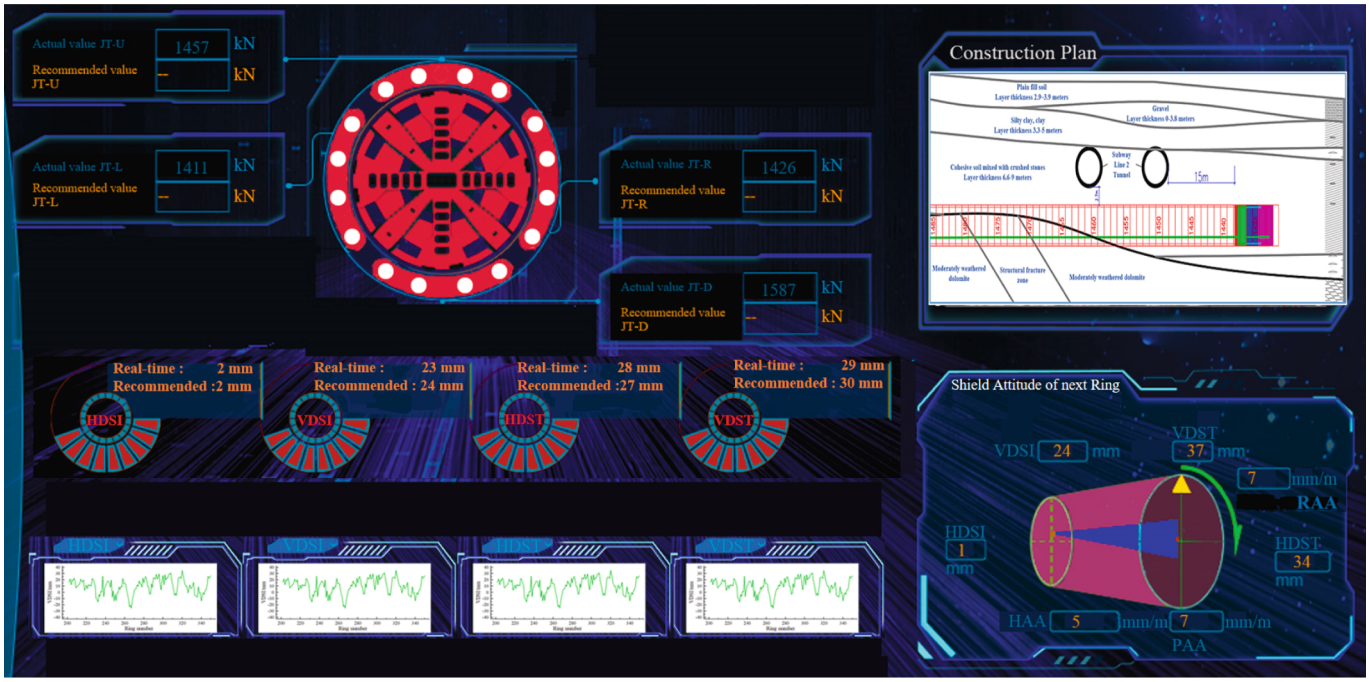
## 5. Conclusions

A hybrid method that integrates IBKA, NGBoost and NSGA-III is proposed for forecasting and optimizing the control of attitude deviation during shield tunneling. A sample repository of shield tunnel construction parameters, geological parameters, and shield tunnel orientation parameters is created on the basis of onsite data collection. The underpass of Line 2 in the Jutang section is used as an example to validate the applicability and efficacy of the proposed method. Through case analysis, the conclusions can be summarized as follows:

(1) The prediction model for shield tunnel attitude parameters that uses IBKA-NGBoost exhibits high precision and efficiency under




(a)




(b)

**Fig. 16.** Interface of the DT platform: (a) Construction parameters and (b) shield attitude.

**Table 9**
Hyperparameter settings for each predictive algorithm.

| Algorithm | Hyperparameter | Setting | Algorithm | Hyperparameter | Setting |
|---|---|---|---|---|---|
| IBKA-BPNN | No. of neurons | 8 | IBKA-GRU | No. of neurons | 20 |
| | Activation function | ReLU | | Epochs | 50 |
| | Learning rate | 0.01 | | Batch size | 20 |
| | Batch size | 200 | | Loss function | MSE |
| | Optimizer | Adam | | Optimizer | Adam |
| IBKA-RF | iterations | 90 | IBKA–XGBoost | iterations | 84 |
| | max_depth | 5 | | colsample_bytree | 0.3 |
| | min_samples_leaf | 1 | | learning_rate | 0.71 |
| | min_samples_split | 2 | | max_depth | 4 |
| NGBoost | Learning_rate | 0.0452 | IBKA-GCN | No. of neurons | 8 |
| | Minibatch_frac | 0.6891 | −LSTM | Activation function | ELU |
| | Col_sampleN_estimators | 0.4788 | | Epochs | 100 |
| | N_estimators | 200 | | Batch size | 32 |
| | – | – | | Loss function | MSE |
| | – | – | | Optimizer | 8 |
| Transformer | Learning rate | 0.003 | | | |
| | Batch size | 64 | | | |
| | Dropout rate | 0.2 | | | |

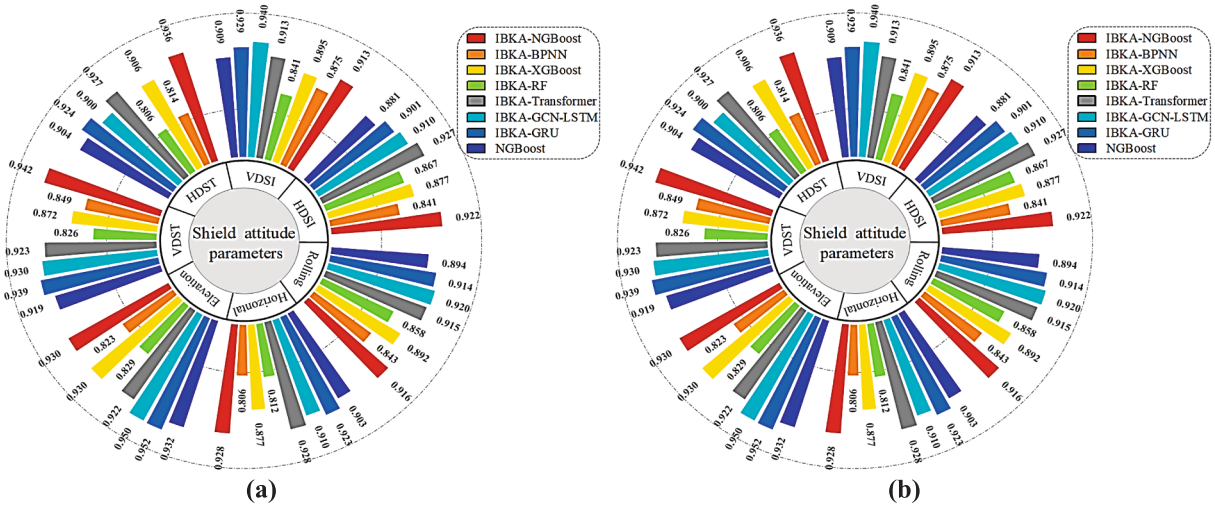



**Fig. 17.** Prediction accuracy of various models: (a) $R^2$ and (b) RMSE.

**Table 10**
Mean training execution time for each predictive algorithm.

| Prediction model | Training time (s) | Prediction model | Training time (s) |
|---|---|---|---|
| IBKA-BPNN | 17.1 | IBKA-GCN-LSTM | 1266.3 |
| IBKA-XGBoost | 15.8 | IBKA-GRU | 685.6 |
| IBKA-RF | 16.5 | NGBoost | 21.6 |
| IBKA-Transformer | 46.9 | IBKA-NGBoost | 29.2 |

**Table 11**
Assessment of prediction accuracy on quartz sandstone test sets.

| Output parameter | $R^2$ | RMSE | MAE |
|---|---|---|---|
| HDSI (F1/mm) | 0.943 | 0.85 | 0.55 |
| VDSI (F2/mm) | 0.910 | 0.95 | 0.71 |
| HDST (F3/mm) | 0.945 | 0.79 | 0.44 |
| VDST (F4/mm) | 0.932 | 0.88 | 0.58 |
| PAA (F5/mm) | 0.941 | 0.82 | 0.48 |
| HAA (F6/mm/m) | 0.952 | 0.65 | 0.39 |
| RAA (F7/mm/m) | 0.916 | 0.89 | 0.67 |

typical geological conditions with small-sample data. Compared with common artificial intelligence algorithm models such as IBKA-BPNN, IBKA-XGBoost, IBKA-RF, IBKA-LSTM, IBKA-GCN-LSTM, NGBoost and IBKA-GRU, this model can achieve the best balance between performance and efficiency. The $R^2$ values for all seven attitude indicators exceed 0.91, the RMSE is less than 0.97, and the MAE is less than 0.71. The training time of IBKA-NGBoost is only 29.2 s.

(2) The IBKA-NGBoost-NSGA-III model notably improves the optimization of multiple objectives, with an average improvement

**Table 12**
Optimization results and improvement percentages of the output parameters for quartz sandstone.

| Output parameter | Avg. of optimization result | Avg. of Raw data | Improvement percentage (%) |
|---|---|---|---|
| HDSI (F1/mm) | 5.88 | 9.88 | 40.5 |
| VDSI (F2/mm) | 9.67 | 18.8 | 48.6 |
| HDST (F3/mm) | 15.56 | 25.38 | 38.7 |
| VDST (F4/mm) | 14.38 | 20.7 | 30.5 |
| PAA (F5/mm) | 5.12 | 9.58 | 46.6 |
| HAA (F6/mm/m) | 1.92 | 3.06 | 37.3 |
| RAA (F7/mm/m) | 2.45 | 2.78 | 11.9 |

**Table 13**
Comparison of artificial methods with DT system decision-making.

| Objective | Avg. DT decision | Avg. the artificial decision | Improvement percentage (%) |
|---|---|---|---|
| HDSI (F1/mm) | 1.29 | 1.64 | 21.10 |
| VDSI (F2/mm) | 2.31 | 3.13 | 26.13 |
| HDST (F3/mm) | 2.53 | 3.55 | 28.81 |
| VDST (F4/mm) | 1.27 | 1.71 | 25.51 |
| PAA (F5/mm) | 1.56 | 2.15 | 27.51 |
| HAA (F6/mm/m) | 0.36 | 0.48 | 25.62 |
| RAA (F7/mm/m) | 0.52 | 0.68 | 23.53 |

percentage of 69.3%. Grounded in the ranking of importance and decision-making of knowledge and experience, by adjusting 12 important construction parameter variables simultaneously, the maximum improvement in controlling the shield tunnel attitude deviation can be achieved, which is 84.8% compared with the original data.

(3) On the basis of the developed IBKA-NGBoost-NSGA-III algorithm and DTs, a shield attitude control system was designed. This system includes an attitude early warning function, a parameter reasoning function and an adaptive regulation and control function, which improve the existing shield regulation and control system.

This research offers a valuable reference for the adaptive regulation and control of shield machine attitudes. However, this study has various limitations and potential challenges. First, in a complex construction environment, the perception ability of DTs is limited, thereby affecting the acquisition of shield operation parameters and the adaptive optimization process. Second, although this study focused on generalization among different geological conditions, the data came from only one engineering case. The performance in projects that involve more complex construction conditions needs to be verified. Furthermore, at present, in practical engineering, the priority of specific target optimization has not been considered, nor have the related physical mechanisms been accounted for. While the suggested algorithm has demonstrated favorable application outcomes in the present study and can also conduct certain importance analyses and provide explanations, the lack of physical mechanisms hinders its promotion and application. Future research should explore the incorporation of physical mechanisms and weights while considering different optimization objectives in the adaptive optimization model. To improve the ability of the model to address the challenges of complexity and ambiguity, the optimal solution that better aligns with the actual engineering requirements that are specified by decision makers should be obtained.

## CRediT authorship contribution statement

**Hongyu Chen:** Writing – original draft, Conceptualization. **Xingchen Zhou:** Formal analysis, Data curation. **Yong Xia:** Resources, Project administration. **Yuan Cao:** Methodology, Investigation. **Yang Liu:** Writing – original draft, Software.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This study received support from the National Natural Science Foundation of China (No. 51778262).

## Data availability

Data will be made available on request.

The findings of this study are supported by data that may be obtained from the corresponding author upon legitimate request.

## References

[1] F. Su, X. Wu, T. Li, Y. Liu, Development of data-driven predictive model and enhanced multiobjective optimization to improve the excavation performance of large-diameter slurry shields, Eng. Appl. Artif. Intel. 162 (2025) 112402.
[2] L. Yang, B. Yu, Y. Liang, Y. Lu, W. Li, Time-varying and non-linear associations between metro ridership and the built environment, Tunn. Undergr. Space Technol. 132 (2023) 104931.
[3] H. Chen, Q. Geoffrey Shen, M.J. Skibniewski, Y. Cao, Y. Liu, Dynamic prediction and optimization of tunneling parameters with high reliability based on a hybrid intelligent algorithm, Inf. Fusion 114 (2025) 102705.
[4] X. Wu, Z. Feng, S. Yang, Y. Qin, H. Chen, Y. Liu, Safety risk perception and control of water inrush during tunnel excavation in karst areas: an improved uncertain information fusion method, Autom. Constr. 163 (2024) 105421.
[5] X. Wu, J. Wang, Z. Feng, H. Chen, T. Li, Y. Liu, Multisource information fusion for real-time prediction and multiobjective optimization of large-diameter slurry shield attitude, Reliab. Eng. Syst. Saf. 250 (2024) 110305.
[6] X. Wu, L. Wang, B. Chen, Z. Feng, Y. Qin, Q. Liu, Y. Liu, Multi-objective optimization of shield construction parameters based on random forests and NSGA-II, Adv. Eng. Inf. 54 (2022).
[7] Y. Cao, S. Li, G.Q. Shen, H. Chen, Y. Liu, Intelligent dynamic control of shield parameters using a hybrid algorithm and digital twin platform, Autom. Constr. 169 (2025) 105882.
[8] H. Chen, Y. Xia, Y. Liu, X. Li, Early warning and prevention of mud cakes in large-diameter slurry shields: Insights from construction parameter data analysis and interpretable machine learning, Reliab. Eng. Syst. Saf 267 (2026) 111886.
[9] H. Huang, J. Chang, D. Zhang, M. Thewes, W. Lin, Improved model-free adaptive control of shield machine attitude during tunnelling, Adv. Eng. Inf. 67 (2025) 103465.
[10] H.Y. Chen, S. Yang, Z.B. Feng, Y. Liu, Y.W. Qin, Safety evaluation of buildings adjacent to shield construction in karst areas: an improved extension cloud approach, Eng. Appl. Artif. Intel. 124 (2023).
[11] Y. Liu, H.Y. Chen, L.M. Zhang, X.J. Wang, Risk prediction and diagnosis of water seepage in operational shield tunnels based on random forest, J. Civ. Eng. Manag. 27 (2021) 539–552.
[12] W. Xu, X. Li, X. Wu, T. Li, H. Chen, Y. Liu, Research on the application of a hybrid intelligent algorithm in reducing the negative response of shield tunneling construction, Eng. Fail. Anal. 169 (2025) 109044.
[13] X. Fu, M. Wu, S. Ponnarasu, L. Zhang, A hybrid deep learning approach for dynamic attitude and position prediction in tunnel construction considering spatio-temporal patterns, Expert Syst. Appl. 212 (2023).
[14] T. Li, J. Liu, X. Wu, F. Su, Y. Liu, Dynamic prediction and control of a tunnel boring machine with a particle swarm optimization–random forest algorithm and an integrated digital twin, Appl. Soft Comput. 178 (2025) 113294.
[15] H. Chen, X. Li, Z. Feng, L. Wang, Y. Qin, M.J. Skibniewski, Z.-S. Chen, Y. Liu, Shield attitude prediction based on Bayesian-LGBM machine learning, Inf. Sci. 632 (2023) 105–129.
[16] W.B. Wang, Y. Zhang, Ieee, research on real-time attitude detection method of shield machine, in: International Conference on Big Data Engineering and Education (BDEE), Guiyang, Peoples R China, 2021, pp. 90–95.
[17] H.J. Liu, J.C. Wang, L.W. Zhang, L. Fei, G.L. Zhao, Ieee, Dynamic Modeling and Trajectory Tracking Control of Tunnel Boring Machine, in: 26th Chinese Control and Decision Conference (CCDC), Changsha, PEOPLES R CHINA, 2014, pp. 4560-4565.
[18] X. Shen, D.J. Yuan, D.L. Jin, Influence of shield attitude change on shield-soil interaction, Appl. Sci.-Basel 9 (2019).
[19] W. Huang, M. Ren, P. Chen, H. Luo, Theoretical analysis model of shield horizontal attitude, Tunnel Construction 42 (2022) 83–89.
[20] L. Wang, X. Yang, G. Gong, J. Du, Pose and trajectory control of shield tunneling machine in complicated stratum, Autom. Constr. 93 (2018) 192–199.
[21] X. Shen, D.J. Yuan, D.L. Jin, X. Wang, X.S. Chen, Shield attitude adjustment induced by slurry pressure balance (SPB) shield tunneling considering the effects of overbreak cutter: a numerical simulation by DEM and engineering application, Urban Rail Transit (2023).
[22] W. Sun, M. Yue, J. Wei, Relationship between rectification moment and angle of shield based on numerical simulation, J. Cent. South Univ. 19 (2012) 517–521.
[23] H. Chen, G.Q. Shen, Z. Feng, Y. Liu, Assessment of the vulnerability of urban metro to rainstorms based on improved cloud model and evidential reasoning, Tunn. Undergr. Space Technol. 157 (2025) 106353.

[24] X. Wu, Z. Feng, J. Liu, H. Chen, Y. Liu, Predicting existing tunnel deformation from adjacent foundation pit construction using hybrid machine learning, Autom. Constr. 165 (2024) 105516.
[25] W. Xu, F.M. Su, J. Liu, X.G. Wu, Y. Liu, Intelligent control and optimization of shield tunneling machines in tunnel construction: Insights from excavation parameter data analysis and interpretable machine learning, Adv. Eng. Inf. 71 (2026).
[26] H. Chen, J. Liu, Q.G. Shen, T. Li, Y. Liu, Data-driven joint multiobjective prediction and optimization for tunnel-induced adjacent bridge pier displacement: A case study in China, Eng. Appl. Artif. Intel. 142 (2025) 109616.
[27] B. Liu, R. Wang, G. Zhao, X. Guo, Y. Wang, J. Li, S. Wang, Prediction of rock mass parameters in the TBM tunnel based on BP neural network integrated simulated annealing algorithm, Tunnelling and Underground Space Technology Incorporating Trenchless Technology Research 95 (2020).
[28] Y. Pan, J. Ni, L. Li, X. Geng, Prediction method of ground surface settlement caused by shield tunnel construction based on LIB - SVM, Journal of Water Resources and Water Engineering 29 (2018) 231–235.
[29] K. Lei, Z. Zhou, H. Li, Z. Tan, L. Gong, A novel time series classification model with Transformer for seamless surrounding rock perception in tunnel boring machine tunneling, Eng. Appl. Artif. Intel. 166 (2026) 113628.
[30] A. Ji, Y. Zhu, H. Fan, X. Xue, M. Zhang, J.X. Zhou, Data-driven transformer-based explainable deep learning model for real-time TBM performance prediction, Reliab. Eng. Syst. Saf. 268 (2026) 111948.
[31] X. Wu, W. Li, T. Li, Z. Feng, F. Su, Y. Liu, A knowledge- and data-driven framework for deformation prediction and control of shield tunneling below existing tunnels, Adv. Eng. Inf. 69 (2026) 103932.
[32] T. Duan, A. Avati, D.Y. Ding, K.K. Thai, S. Basu, A. Ng, A. Schuler, S. Assoc Informat, NGBoost: Natural Gradient Boosting for Probabilistic Prediction, in: 25th Americas Conference on Information Systems of the Association-for-Information-Systems(AMCIS), Cancun, MEXICO, 2019.
[33] Y. Cao, F.M. Su, M.F. Antwi-Afari, J. Lei, X.G. Wu, Y. Liu, Enhancing mix proportion design of low carbon concrete for shield segment using a combination of Bayesian optimization-NGBoost and NSGA-III algorithm, J. Clean. Prod. 465 (2024).
[34] V.V. Degtyarev, S.J. Hicks, Machine learning-based probabilistic predictions of shear resistance of welded studs in deck slab ribs transverse to beams, Steel and Composite Structures 49 (2023) 109–123.
[35] J.Y. Ding, D.C. Feng, E. Brunesi, F. Parisi, G. Wu, Efficient seismic fragility analysis method utilizing ground motion clustering and probabilistic machine learning, Eng. Struct. 294 (2023).
[36] T. Kavzoglu, A. Teke, Predictive performances of ensemble machine learning algorithms in landslide susceptibility mapping using random forest, extreme gradient boosting XGBoost) and Natural Gradient Boosting (NGBoost), Arab. J. Sci. Eng. 47 (2022) 7367–7385.
[37] S. Liu, X. Li, R. Ren, W. Liu, Z. Feng, L. Zhang, J. Liu, H. Chen, X. Wu, Method for predicting deformation of adjacent existing tunnel caused by foundation pit construction based on Bayesian optimization-Natural Gradient Boosting-SHapley Additive exPlanations, involves introducing SHapley addition interpretation method to perform interpretation analysis on model, in, Univ Huazhong Sci & Technology (Uyhz-C).
[38] M. Zhou, P. Li, Y. Qin, W. Xu, W. Liu, Q. Zhong, H. Zhu, Y. Hu, F. Lin, H. Zhang, H. Chen, X. Wu, Shield proximity structure safety predictive control method based on intelligent algorithm, involves outputting corresponding safety control strategy according to safety risk level of shield proximity structure, in, Univ Huazhong Sci & Technology (Uyhz-C).
[39] Y. Zhang, L. Liu, J. Wu, S. Zeng, J. Hu, Y. Tao, Y. Huang, X. Zhou, X. Liang, An interpretable probabilistic prediction algorithm for shield movement performance, Front. Earth Sci. 12 (2024).
[40] L. Zhang, J. Guo, X. Fu, R.L.K. Tiong, P. Zhang, Digital twin enabled real-time advanced control of TBM operation using deep learning methods, Autom. Constr. 158 (2024) 105240.
[41] D. Chakraborty, H. Elhegazy, H. Elzarka, L. Gutierrez, A novel construction cost prediction model using hybrid natural and light gradient boosting, Adv. Eng. Inf. 46 (2020) 101201.
[42] L. Yang, H. Yang, B. Yu, Y. Lu, J. Cui, D. Lin, Exploring non-linear and synergistic effects of green spaces on active travel using crowdsourced data and interpretable machine learning, Travel Behav. Soc. 34 (2024) 100673.
[43] Z. Cai, Y. Long, L. Shao, Classification complexity assessment for hyper-parameter optimization, Pattern Recogn. Lett. 125 (2019).
[44] L. Yang, H. Yang, J. Tang, Y. Lu, J. Liu, Assessing attribute performance and older adults' satisfaction with transit services: implications for age-friendly planning, Travel Behav. Soc. 40 (2025) 101026.
[45] F. Yan, Z. Lin, X. Wang, F. Azarmi, K. Sobolev, Evaluation and prediction of bond strength of GFRP-bar reinforced concrete using artificial neural network optimized with genetic algorithm, Compos. Struct. 161 (2017).
[46] A. Jiao, Z. Jiao, B. Gao, Data-driven seismic mechanical performance evaluation of RC columns based on adaptive optimization ensemble learning method integrating random forest and back propagation neural network, Results Eng. 27 (2025) 105810.
[47] K. Deb, H. Jain, An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: solving problems with box constraints, IEEE Trans. Evol. Comput. 18 (2014).
[48] K. Deb, H. Jain, An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: solving problems with box constraints, ieee transactions on, Evol. Comput. 18 (2014) 577–601.
[49] F.W. Ge, K. Li, Y. Han, Solving interval many-objective optimization problems by combination of NSGA-III and a local fruit fly optimization algorithm, Appl. Soft Comput. 114 (2022).
[50] L. She, C.-C. Hu, Y.-L. Li, Z.-Y. Li, Q. Song, Y. Zhang, M.-M. He, Intelligent decision of TBM operating parameters: a multi-objective optimization approach based on tabular deep learning, Adv. Eng. Inf. 68 (2025) 103573.
[51] J. Li, L. Jing, X. Zheng, P. Li, C. Yang, Application and outlook of information and intelligence technology for safe and efficient TBM construction, Tunn. Undergr. Space Technol. 93 (2019) 103097.
[52] J. Barata, I. Kayser, How will the digital twin shape the future of industry 5.0? Technovation 134 (2024) 103025.
[53] G. 50446-, Code for construction and acceptance of shield tunneling method, in, Ministry of Housing and Urban-Rural Construction of the People's Republic of …, 2017.
[54] S. Feng, Z. Chen, H. Luo, S. Wang, Y. Zhao, L. Liu, D. Ling, L.J.T. Jing, U.S. Technology, Tunnel boring machines (TBM) performance prediction: A case study using big data and deep learning, 110 (2021) 103636.
[55] L. Li, Z. Liu, J. Shen, F. Wang, W. Qi, S. Jeon, A LightGBM-based strategy to predict tunnel rockmass class from TBM construction data for building control, Adv. Eng. Inf. 58 (2023) 102130.
[56] W. Luo, W. Liu, W. Liu, L. Xia, J. Zheng, Y. Liu, Analysis of influencing factors and carbon emission scenario prediction during building operation stage, Energy 316 (2025) 134401.
[57] J. Wang, W.-C. Wang, X.-X. Hu, L. Qiu, H.-F. Zang, Black-winged kite algorithm: a nature-inspired meta-heuristic for solving benchmark functions and engineering problems, Artif. Intell. Rev. 57 (2024) 98.
[58] S. Huang, Y. Xing, J. Cao, X. Zhang, Y. Wang, Identification of optimal parameters of PEMFC steady-state model using improved black kite algorithm, Int. J. Hydrogen Energy 106 (2025) 1302–1321.
[59] S.-I. Amari, Natural gradient works efficiently in learning, Neural Comput. 10 (1998) 251–276.
[60] M.C. Iban, S.S. Bilgilioglu, Snow avalanche susceptibility mapping using novel tree-based machine learning algorithms (XGBoost, NGBoost, and LightGBM) with eXplainable Artificial Intelligence (XAI) approach, Stochastic Environmental Research and Risk Assessment, (2023).
[61] H.Y. Chen, J.Y. Wang, Z.B. Feng, Y. Liu, W. Xu, Y.W. Qin, Research on the risk evaluation of urban wastewater treatment projects based on an improved fuzzy cognitive map, Sustain. Cities Soc. 98 (2023).
[62] Y. Liu, H. Chen, L. Zhang, X. Wu, X.-J. Wang, Energy consumption prediction and diagnosis of public buildings based on support vector machine learning: a case study in China, J. Clean. Prod. 272 (2020) 122542.
[63] L.-Y. Huang, X. Zhang, Q. Wang, Z.-S. Chen, Y. Liu, Evaluating media knowledge capabilities of intelligent search dialogue systems: a case study of ChatGPT and new bing, J. Knowl. Econ. (2024).
[64] L. Yang, C. Hongyu, Research on factors influencing total carbon emissions of construction based on structural equation modeling: A case study from China, Building and Environment 275 (2025) 112396.
[65] X. Wu, X. Li, Y. Qin, W. Xu, Y. Liu, Intelligent multiobjective optimization design for NZEBs in China: four climatic regions, Appl. Energy 339 (2023) 120934.
[66] H. Chen, Y. Cheng, T. Du, X. Wu, Y. Cao, Y. Liu, Enhancing the performance of recycled aggregate green concrete via a Bayesian optimization light gradient boosting machine and the nondominated sorting genetic algorithm-III, Constr. Build. Mater. 458 (2025) 139527.
[67] Y. Chen, M. Wang, Y. Zhang, Y. Lu, B. Xu, L. Yu, Cascade hydropower system operation considering ecological flow based on different multi-objective genetic algorithms, Water Resour. Manag. 37 (2023) 3093–3110.
[68] I. Khettabi, L. Benyoucef, M.A. Boutiche, Sustainable multi-objective process planning in reconfigurable manufacturing environment: adapted new dynamic NSGA-II vs New NSGA-III, Int. J. Prod. Res. 60 (2022) 6329–6349.
[69] W. Xu, X. Wu, S. Xiong, T. Li, Y. Liu, Optimizing the sustainable performance of public buildings: A hybrid machine learning algorithm, Energy 320 (2025) 135283.
[70] K. Zhang, Z. Min, W. Huang, T.F.P. Henning, X. Hao, K. Shao, Climate-resilient epoxy asphalt mixture design: an intelligent framework, Adv. Eng. Inf. 65 (2025) 103395.
[71] J. Huang, A. Ji, L. Zhang, X. Fu, X. Song, Data-driven optimization for enhanced excavation efficiency in tunnel construction: a case study, Eng. Appl. Artif. Intel. 142 (2025).
[72] Y. Liu, H. Chen, X.-j. Wang, Research on green renovations of existing public buildings based on a cloud model –TOPSIS method, J. Build. Eng 34 (2021) 101930.
[73] L. Yang, Y. Ao, J. Ke, Y. Lu, Y. Liang, To walk or not to walk? Examining non-linear effects of streetscape greenery on walking propensity of older adults, J. Transp. Geogr. 94 (2021) 103099.
[74] L. Yang, Integrating six feeder modes with the metro system: insights from on-demand and pooled services, Transp. Res. Part E Logist. Transp. Rev. 208 (2026) 104659.
[75] C.-H. Wu, G.-H. Tzeng, R.-H.J.E.S.w.A. Lin, A novel hybrid genetic algorithm for kernel function and parameter optimization in support vector regression, 36 (2009) 4725-4735.
[76] J. Xu, Z. Zhang, L. Zhang, D. Liu, Predicting shield position deviation based on double-path hybrid deep neural networks, Autom. Constr. 148 (2023).
[77] Z. Dai, P. Li, M. Zhu, H. Zhu, J. Liu, Y. Zhai, J. Fan, Dynamic prediction for attitude and position of shield machine in tunneling: a hybrid deep learning method considering dual attention, Adv. Eng. Inf. 57 (2023).

[78] X. Fu, S. Ponnarasu, L. Zhang, R.L.K. Tiong, Online multi-objective optimization for real-time TBM attitude control with spatio-temporal deep learning model, Autom. Constr. 158 (2024).
[79] H. Xiao, B. Xing, Y. Wang, P. Yu, L. Liu, R. Cao, Prediction of shield machine attitude based on various artificial intelligence technologies, Appl. Sci. 11 (2021).
[80] H. Sun, S. Yang, Range-free localization algorithm based on modified distance and improved black-winged kite algorithm, Comput. Netw. 259 (2025) 111091.